ELLIPTIC TWISTS AND DEHN TWISTS ON THE ONE POINT KÄHLER BLOWUP OF THE K3 SURFACE

YI DU, TIAN-JUN LI

Abstract. Let X be the K3 surface and $\tilde{X} = X\#\overline{\mathbb{CP}}^2$. Let $\mathcal{KS}(\tilde{X})$ be the space of Kähler type forms on $\tilde{X}$ and $K(\tilde{X}, \tilde{\omega})$ the smoothly trivial symplectic mapping class group for $\tilde{\omega} \in \mathcal{KS}(\tilde{X})$. By generalizing the techniques in [40] and [39], we show that, for a large subspace of $\mathcal{KS}(\tilde{X})$ that contains an open and dense subset, $K(\tilde{X}, \tilde{\omega})$ is nontrivial and infinitely generated. We speculate that this is true for all Kähler type forms (even all symplectic forms) on $\tilde{X}$. The nontrivial mapping classes we detect arise from either small deformations of elliptic Hodge type twists or spherical Hodge type twists.

Moreover, we show that elliptic Hodge type twists are not generated by squared Seidel-Dehn twists. Thus, we obtain example of simply connected symplectic 4-manifold whose smoothly trivial symplectic mapping class group is not generated by squared Seidel-Dehn twists. This phenomenon complements the examples in [23] concerning Donaldson's question whether the Torelli symplectic mapping class group is generated by squared Seidel-Dehn twists.

Contents

1. Introduction 2
1.1. The space of Kahler type forms on $\tilde{X}$ and $Se(\kappa)$ 2
1.2. Hodge type twists and small deformations 3
1.3. The spherical and elliptic q homomorphisms q_S and q_T 4
2. Hodge type twists 6
2.1. Spherical Hodge type twists and squared Seidel-Dehn twists 6
2.2. Elliptic Hodge type twists and elliptic twists 8
3. Spherical and elliptic Hodge type twists on $(\tilde{X}, \tilde{\omega})$ 9
3.1. Kähler geometry of (X, j) and the period domain Ω 9
3.2. The geometry of (X, κ) 11
3.3. The index sets $S(X, \kappa)$ and $T_\lambda(X, \kappa)$ 14
3.4. Constructing Hodge type twists 16
4. The Q invariant for $\mathcal{A}_\omega$-noneffective ϵ, q_S and q_T 20
4.1. Relative family Seiberg–Witten invariants in the canonical chamber 20
4.2. The generalized invariant $Q_\epsilon : \pi_1(\mathcal{S}_\omega) \to \mathbb{Z}$ for $\mathcal{A}_\omega$-noneffective ϵ 24
4.3. The spherical and elliptic q homomorphisms on $K(\tilde{X}, \tilde{\omega})$ 27
References 30

1. Introduction

Let (M,ω) be a closed symplectic 4-manifold and $\mathrm{Symp}(M,\omega)$ its symplectomorphism group. Consider the symplectic mapping class group $\pi_0\mathrm{Symp}(M,\omega)$ and denote it by $SMC(M,\omega)$. There are two important subgroups of $\mathrm{Symp}(M,\omega)$: the group of homologically trivial symplectomorphisms $\mathrm{Symp}_h(M,\omega)$, and the group of smoothly trivial symplectomorphisms

$$\mathrm{Symp}_{st}(M,\omega) = \mathrm{Symp}(M,\omega) \cap \mathrm{Diff}_0(M).$$

Denote the Torelli Symplectic mapping class group $\pi_0(\mathrm{Symp}_h(M,\omega))$ by $TSMC(M,\omega)$ and the smoothly trivial symplectic mapping class group $\pi_0(\mathrm{Symp}_{st}(M,\omega))$ by $K(M,\omega)$.

Question 1.1. Is there a closed symplectic 4-manifold M such that $K(M,\omega)$ is nontrivial for any symplectic form ω on M?

Although several constructions have appeared, this question and even the weaker versions about SMC and $TSMC$ remain open. Let X denote the K3 surface, which is the closed, oriented, simply connected 4-manifold admitting complex structures with vanishing canonical class. It is shown that $K(X,\omega)$ is infinitely generated for certain nongeneric families of Kähler type forms in [36] and [40]. Let $\tilde{X} = X\#\overline{\mathbb{CP}}^2$, then $\tilde{X}$ is also an oriented, simply connected 4-manifold admitting complex structures. Let e denote a generator of $H^2(\overline{\mathbb{CP}}^2;\mathbb{Z})$ (as well as its Poincaré dual). For a symplectic form $\tilde{\omega}$ on $\tilde{X}$ let $\lambda(\tilde{\omega}) = |\tilde{\omega}(e)|$. $\lambda(\tilde{\omega})$ is necessarily positive. With respect to the natural projection $p_* : H^2(\tilde{X};\mathbb{R}) = H^2(X;\mathbb{R}) \oplus \mathbb{R}e \to H^2(X;\mathbb{R})$, let $\kappa(\tilde{\omega}) = p_*[\tilde{\omega}] \in H^2(X;\mathbb{R})$.

In this paper, we investigate $K(\tilde{X},\tilde{\omega})$ when $\tilde{\omega}$ is of Kähler type and indicate that $\tilde{X}$ may provide a positive answer to Question 1.1.

1.1. The space of Kahler type forms on $\tilde{X}$ and $Se(\kappa)$.

Following [15], we say that a symplectic form on M is of Kähler type if it is compatible with some complex structure. The symplectic geometry of Kähler type forms has been studied in [14][15][40]. Let $\mathcal{S}(M)$ be the space of symplectic forms and $\mathcal{KS}(M)$ the space of Kähler type forms on M.

For the K3 surface X, $\mathcal{KS}$ is open in $\mathcal{S}(X)$ [41][15], and it is conjectured by Donaldson that $\mathcal{KS}(X) = \mathcal{S}(X)$. Cohomologous Kähler type forms on X are diffeomorphic [43][15].

For a Kähler type form ω on X, $(\tilde{X},\tilde{\omega})$ is called a Kähler type blowup of (X,ω) if $\tilde{\omega}$ is compatible with some complex structure $\tilde{J}$ on $\tilde{X}$ such that $(\tilde{X},\tilde{\omega},\tilde{J})$ arises from a holomorphic ball embedding of size λ into a Kähler tuple (X,ω,J). And the size of $\tilde{\omega}$ is defined as $\lambda = \lambda(\tilde{\omega})$. In particular, every Kähler type $\tilde{\omega}$ is a Kähler type blowup form.

The space of Kähler type forms on $\tilde{X}$ is investigated in [30].

Theorem 1.2. *([30]) Cohomologous Kähler type forms on $\tilde{X}$ are diffeomorphic. And $\mathcal{KS}(\tilde{X})$ is open in $\mathcal{S}(\tilde{X})$.*

In particular, different conected components of $\mathcal{KS}(\tilde{X})$ are diffeomorphic, so we use the $\mathcal{KS}(\tilde{X})$ to denote both the space of Kähler type forms and a connected component of it when no confusion arises.

We next briefly introduce the κ-Seshadri constant function on X (Definition 3.12), $\mathrm{Se} : V^+(X) \to \mathbb{R}^{\geqslant 0}$, where $V^+(X) \subset H^2(X;\mathbb{R})$ is the cone of positive square classes. To do this, we need to introduce $\mathcal{M}_\kappa^+$.

Definition 1.3. For any $\kappa \in V^+(X)$, let $\mathcal{M}_\kappa^+$ be the κ-polarized moduli space of complex K3 surfaces. We say that $u \in \mathcal{M}_\kappa^+$ is generic if the Neron-Severi group of X_u satisfies

$$NS(X_u) = H^{1,1}(X_u) \cap H^2(X,\mathbb{Z}) = \mathbb{R}\kappa \cap H^2(X,\mathbb{Z}).$$

We use $\mathcal{M}_\kappa^{gen}$ to denote the set of generic points in $\mathcal{M}_\kappa^+$.

It is easy to see that $\mathcal{M}_\kappa^{gen}$ is the complement of a countable union of hyperplanes and hence is dense in $\mathcal{M}_\kappa^+$. The κ-Seshadri constant function $\mathrm{Se}(\kappa)$ is the sharp lower bound of the size of maximal Kähler ball embeddings over Kähler K3 surfaces parametrized by $\mathcal{M}_\kappa^{gen}$. When κ is a primitive integral class, $\mathrm{Se}(\kappa)$ is also the lower bound of ordinary Seshadri constants for $u \in \mathcal{M}_\kappa^{gen}$. Clearly, $\mathrm{Se}(\kappa) \leqslant \sqrt{\kappa^2}$. We will show in Section 3

Lemma 1.4. *For a class $\kappa \in V^+(X)$,*

- *[14] there is a Kähler type form on $\tilde{X}$ in the class $\kappa - \lambda e$ if $\lambda \in (0, Se(\kappa))$.*
- *[14] $\mathrm{Se}(\kappa) = \sqrt{\kappa^2}$ when κ is not a multiple of any integral class.*
- *When $\lambda < \mathrm{Se}(\kappa)$, $\mathcal{M}_{\kappa,\lambda}^+ \supset \mathcal{M}_\kappa^{gen}$. Here $\mathcal{M}_{\kappa,\lambda}^+$ is the subspace of $\mathcal{M}_\kappa^+$ parameterizing complex structures that admit holomorphic balls of size λ centered at any point.*
- *$\mathrm{Se}(\kappa) \geqslant \frac{\sqrt{2}}{2}\sqrt{\kappa^2}$.*

1.2. **Hodge type twists and small deformations.** We next briefly discuss Hodge type twists for Kähler surfaces with $p_g \geqslant 1$. To motivate, we recall Kronheimer's fibration in [26]. For any symplectic manifold (M,θ) we have a fibration

$$\mathrm{Symp}_{st}(M,\theta) \xrightarrow{i} \mathrm{Diff}_0(M) \xrightarrow{p} \mathcal{S}_\theta, \tag{1.1}$$

and the associated long exact sequence

$$\cdots \to \pi_1(\mathrm{Diff}_0(M)) \xrightarrow{p_*} \pi_1(\mathcal{S}_\theta,\theta) \xrightarrow{\partial_*} K(M,\theta) = \pi_0(\mathrm{Symp}_{st}(M,\theta)) \to 1 = \pi_0(\mathrm{Diff}_0(M)). \tag{1.2}$$

Here, $\mathcal{S}_\theta$ denotes the space of symplectic forms isotopic to θ. It follows from (1.2) that we have a surjection

$$\pi_1(\mathcal{S}_\theta,\theta) \to K(M,\theta).$$

When θ is a Kahler type form in M with $b^+(M) \geqslant 3$, Hodge type twists are mapping classes from certain kinds of loops in $\mathcal{S}_\theta$. Note that $\mathcal{S}_\theta \subset \mathcal{KS}(M)$ by (1.1). To construct Hodge type twists on (M,θ) with canonical class K, we pick a class ϵ with $d(\epsilon) = \epsilon^2 + K\cdot\epsilon = -2$ and $\langle\theta,\epsilon\rangle \leqslant 0$. Then a Hodge type twist is defined by a loop ℓ in $\mathcal{S}_\theta$ that bounds an admissible disk in the larger space $\mathcal{KS}(M)$ satisfying all the conditions in Definition 2.1. The spherical Hodge type twists from ϵ with $\epsilon^2 = -2$ are based on [40] and the elliptic Hodge type twists from ϵ with $\epsilon^2 = -1$ are based on [39].

In Section 2, we relate these two Hodge type twists to the squared Seidel-Dehn twists and the elliptic twists studied in [37]. An elliptic Hodge type twist is an elliptic twist, while a suitable small deformation of elliptic twists in $\mathcal{S}$ along a path of cohomologous loops

remains an elliptic twist. While spherical Hodge type twists and squared Seidel-Dehn twists are clearly closely related, the precise relation remains an interesting question.

In section 3, we restrict to $\tilde{X}$ and construct abundant spherical and elliptic Hodge type twists for Kähler type forms $\tilde{\omega}$ subject to the mild condition $\lambda(\tilde{\omega}) < \mathrm{Se}(\kappa(\tilde{\omega}))$. For any κ and $\lambda \in \big(0, \mathrm{Se}(\kappa)\big)$, we pick appropriate loops in the sub-domain $\mathcal{M}^+_{\kappa,\lambda}$ of $\mathcal{M}^+_\kappa$ to construct spherical and elliptic Hodge type twists on $\tilde{X}$. For these loops in $\mathcal{M}^+_{\kappa,\lambda}$ we use Lemma 1.4 to get appropriate loops in $\mathcal{KS}_{\kappa-\lambda e}(\tilde{X})$ and then extend such loops to appropriate disks in $\mathcal{KS}(\tilde{X})$, defining the Hodge type twists from loops in $\mathcal{M}^+_{\kappa,\lambda}$. Here, $\mathcal{KS}_{\kappa-\lambda e}(\tilde{X})$ denotes the space of Kähler type forms in the class $\kappa - \lambda e$. Let

$$\bar{S}(X,\kappa) := \{\gamma \in H^2(X;\mathbb{Z}) | \gamma^2 = -2, \langle \kappa, \gamma\rangle = 0\}/\gamma \sim -\gamma, \tag{1.3}$$

$$T_\lambda(X,\kappa) := \{\delta \in H^2(X;\mathbb{Z})^{prim} | \delta^2 = 0, 0 < \langle \kappa, \delta\rangle < \lambda\}, \tag{1.4}$$

where $H^2(X;\mathbb{Z})^{prim}$ is the set of primitive classes. Let $ST(X,\kappa,\lambda) = \bar{S}(X,\kappa) \cup T_\lambda(X,\kappa)$.

Lemma 1.5. *$ST(X,\kappa,\lambda)$ is nonempty, and in fact an infinite set.*

(1) $\bar{S}(X,\kappa)$ is infinite when κ is a real multiple of an integral class.

(2) $T_\lambda(X,\kappa)$ is infinite when κ is not a multiple of an integral class.

Let $U_{TH}, U_{SH} \subset \mathcal{KS}(\tilde{X})$ be subspaces where there is an elliptic Hodge type twist, a spherical Hodge type twist in $K(\tilde{X},\tilde{\omega})$ respectively.

Theorem 1.6. *If $\lambda(\tilde{\omega}) < \mathrm{Se}(\kappa(\tilde{\omega}))$, there are spherical and elliptic Hodge type twists on $(\tilde{X},\tilde{\omega})$ indexed respectively by $\bar{S}(X,\kappa(\tilde{\omega}))$ and $T_{\lambda(\tilde{\omega})}(X,\kappa(\tilde{\omega}))$. Consequently, U_{TH} contains the connected dense subspace of $\mathcal{KS}(\tilde{X})$ where $\kappa(\tilde{\omega})$ is not a multiple of an integral class.*

1.3. **The spherical and elliptic q homomorphisms q_S and q_T.** Kronheimer defines a Seiberg–Witten theoretic homomorphism $Q^\theta_\epsilon : \pi_1(\mathcal{S}_\theta,\theta) \to \mathbb{Z}$ for a symplectic 4-manifold M with $b^+ > 3$ and a class ϵ with $d(\epsilon) = -2$ and $\langle\theta,\epsilon\rangle \leqslant 0$ [26]. We need two extensions: (1), M is allowed to have $b^+ \leqslant 3$ by choosing a canonical ω chamber in [40] and [33]; (2), ϵ is assumed to be $\mathcal{A}_\omega$-noneffective (which could have a positive pairing with θ).

Assume that both Q_ϵ and $Q_{K-\epsilon}$ are defined, their difference $q^\theta_\epsilon := Q_\epsilon - Q_{K-\epsilon}$ vanishes on $\mathrm{Im}\big(\pi_1(\mathrm{Diff}_0(\tilde{X})) \to \pi_1(\mathcal{S}_\theta,\theta)\big)$ as in [40] [39]. Thus, there is a homomorphism via (1.2),

$$q^\theta_\epsilon : K(M,\theta) \to K^{ab}(M,\theta) \to \mathbb{Z},$$

where $K^{ab}(M,\theta)$ is the abelianization of $K(\tilde{X},\tilde{\omega})$. For our purposes, we use the simpler $\mathbb{Z}_2$ valued invariants.

For symplectic forms on $\tilde{X}$, the two extensions of Q allow us to define abundant q_ϵ.

Theorem 1.7. *For a symplectic form $\tilde{\omega}$ on $\tilde{X}$, there are spherical and elliptic q invariants, q_γ and $q_{\delta-e}$, indexed by $\overline{S}(X,\kappa(\tilde{\omega}))$ and $T_{\lambda(\tilde{\omega})}(\tilde{X},\kappa(\tilde{\omega}))$ respectively. Moreover,*

(1) (stability) $q_{\delta-e}$ is defined in a neighborhood of $\tilde{\omega}$ in $\mathcal{S}(\tilde{X})$, and if it is nontrivial it remain nontrivial in a possibly smaller neighborhood.

(2) $q_{\delta-e}$ vanishes on any squared Seidel-Dehn twist.

(3) if $\tilde{\omega}$ is of Kähler type, $q_S = \oplus q_\gamma$ is nontrivial on each spherical Hodge type twist, and $q_T = \oplus q_{\delta-e}$ is nontrivial on each elliptic Hodge type twist.

(4) if $\tilde{\omega}$ is of Kähler type and $\lambda(\tilde{\omega}) < Se(\kappa(\tilde{\omega}))$, then $q_S \oplus q_T$ is surjective.

Consequently, $\tilde{X}$ potentially provides a solution to Question 1.1. Let $U(\tilde{X}) = \{\tilde{\omega} \in \mathcal{KS}(\tilde{X}) | K(\tilde{X}, \tilde{\omega}) \neq \{1\}\}$. We have the following partial answer to Question 1.1 from Theorem 1.6 and (1) of Theorem 1.7, and Theorem 1.2.

Corollary 1.8. *$U(\tilde{X})$ contains an connected, open and dense subset of $\mathcal{KS}(\tilde{X})$, and an open subset of $\mathcal{S}(\tilde{X})$.*

Proof. It follows from Theorem 1.6 and the first and third results in Theorem 1.7. □

Let U_∞ be the space of symplectic forms $\tilde{\omega}$ where $K(\tilde{X}, \tilde{\omega})$ is infinitely generated. We will also use (4) of Theorem 1.7 to show that $U_\infty(\tilde{X})$ also contains an open and dense subset of $\mathcal{KS}(\tilde{X})$ theorem 4.26.

Corollary 1.8 is related to the following question by Donaldson in [36].

Question 1.9. If (M, ω) is a simply connected and closed symplectic 4-manifold, do we have $DS(M, \omega) = TSMC(M, \omega)$?

Here, $DS(M, \omega)$ is the subgroup of $SMC(M, \omega)$ generated by the squared Seidel-Dehn twist τ_L^2 for a Lagrangian sphere $L \subset (M, \omega)$. By [Lemma 2.5 and Proposition 2.3, [35]], the square of τ_L is isotopic to the identity smoothly. Therefore, the group $K(M, \omega)$ fits into the inclusions

$$DS(M, \omega) \subset K(M, \omega) \subset TSMC(M, \omega).$$

By [Theorem 0, [27]], $K(M, \omega) = TSMC(M, \omega)$ for rational surfaces. And by [Theorem 1.6, [28]], $K(M, \omega) = DS(M, \omega)$ for positive rational surfaces.

The first family of negative examples for Question 1.9 appeared in [23], where it is shown that $K \neq TSMC$ for infinitely many minimal elliptic surfaces. We get a different kind of negative answer to Question 1.9 from Theorem 1.7 (4).

Corollary 1.10. *The set $V' = \{\tilde{\omega} \in \mathcal{KS}(\tilde{X}) | DS(\tilde{X}, \tilde{\omega}) \neq TSMC(\tilde{X}, \tilde{\omega})\}$ contains U_{TH}. Moreover, for each $\tilde{\omega} \in U_{TH}$, there is a neighborhood $U_{\tilde{\omega}} \subset \mathcal{KS}(\tilde{X})$ of $\tilde{\omega}$ contained in $U_{TH} \subset V'$.*

Proof. By the third item in Theorem 1.7, for each $\tilde{\omega} \in U_{TH}$, $q_{\delta - e}$ is non-trivial. By the first item in Theorem 1.7, $q_{\delta - e}$ is non-trivial for a neighborhood $U_{\tilde{\omega}}$. By the second item in Theorem 1.7, $DS(\tilde{X}, \tilde{\omega}') \neq TSMC(\tilde{X}, \tilde{\omega}')$ for each $\tilde{\omega}' \in U_{\tilde{\omega}}$. □

The organization of this paper is as follows. In Section 2 we define Hodge type twists and relate them to various types of mapping classes. In Section 3 we review the Kähler geometry of the K3 surface and prove Lemma 1.5, Lemma 1.4, and Theorem 1.6. In Section 4 we review the family Seiberg–Witten theory and introduce the generalized Q invariant and the associated q invariant. We prove Theorem 1.7.

Acknowledgement We'd like to thank Jonny Evans, Gleb Smirnov and Jianfeng Lin for very helpful communication. We thank Hokuto Konno, Jun Li, Shengzhen Ning, Weiwei Wu and Guangbo Xu for useful discussions. We are partially supported by Simons Travel Fund SFI-MPS-TSM-00013390.

AI use. The authors don't use AI in this paper.

2. Hodge type twists

Several constructions of Torelli symplectic mapping classes have appeared. Potential nontrivial Torelli symplectic mapping classes include the squared Seidel Dehn twist τ_L along Lagrangian (-2) spheres [34], elliptic twist defined by a symplectic (-1) torus and suitable symplectic deformation [37][39][7] and Lagrangian $\mathbb{RP}^2$ twist along Lagrangian $\mathbb{RP}^2$ [17][1], ball swappings in [44]. Also, there is some recent work about smoothly non-trivial fibered Dehn twist on the contact boundary of symplectic fillings in dimension 4 [25][24].

Now we suppose that M is a Kähler surface, which means that it admits Kähler type forms. We will define the "Hodge type" twist of a Kähler type symplectic form ω.

Definition 2.1. Let $\epsilon \in H^2(M;\mathbb{Z})$ with $d(\epsilon) := \frac{\epsilon^2 - \langle K_\omega, \epsilon\rangle}{2}$. An ϵ-Hodge type twist ϕ_ϵ is the monodromy of a loop of Kähler type forms $\ell := \{\omega_t | t \in \partial\mathbb{D}^2 \subset \mathbb{C}\} \subset \mathcal{S}_\omega$ based at $\omega = \omega_1$ such that

- There is a holomorphic disc family D of Kähler structures $\{(\omega_z, J_z)|z \in \mathbb{D}^2\}$ such that $\{\omega_z\}$ extends ℓ,
- $\langle [\omega], \epsilon\rangle \leqslant 0$,
- $[\omega_z]\cdot[\omega] > 0, \forall z \in \mathbb{D}^2$,
- $\epsilon \notin H^{1,1}(M, J_z;\mathbb{R})$, $\forall z \neq 0$,
- PD(ϵ) is represented by a smooth curve in (M, J_0), ϵ is not in the kernel of $\Omega_*(\partial_t)$ and the linear system has complex dimension $d(\epsilon)+1$.

Here, $\Omega_* : T_bD \to \mathrm{Hom}(H^{1,1}, H^{2,0})$ is the infinitesimal variation of Hodge structures and ∂_t is a generator of T_bD. As observed in [26][40], suppose the moduli space of ϵ curves is discrete and the curves are smooth, then it is transversely cut if $\epsilon \in H^{1,1}$ is not in the kernel of $\Omega_*(\partial_t)$ at any point $b \in D$ supporting an ϵ curve.

In this paper, we study two special cases of Hodge type twists.

Definition 2.2. If $\epsilon^2 = -2$ and $\langle K_\omega, \epsilon\rangle = 0$, then the ϵ-Hodge type twist is called a spherical Hodge type twist. If $\epsilon^2 = -1$ and $\langle K_\omega, \epsilon\rangle = 1$, then the ϵ-Hodge type twist is called an elliptic Hodge type twist.

In this section, we study some properties of the spherical and elliptic Hodge type twists.

2.1. Spherical Hodge type twists and squared Seidel-Dehn twists.

Let (M,ω) be a closed symplectic 4-manifold and let $L \subset M$ be a Lagrangian sphere.

Let T^*S^2 be represented by $\{(u,v) \in \mathbb{R}^3\times\mathbb{R}^3 | \langle u, v\rangle = 0, ||v|| = 1\}$ and let $\omega = du\wedge dv$. There is a Hamiltonian circle action on $T^*S^2\backslash S^2$ with moment map $h(u,v) = ||u||$:

$$\sigma_t(u,v) = (\cos(t)u - \sin(t)||u||v, \quad \cos(t)v + \sin(t)\frac{u}{||u||}), t \in [0, 2\pi].$$

Notice that $\sigma_\pi = -id$. Now take a function $r : \mathbb{R} \to \mathbb{R}$ satisfying $r(t) = 0$ for $t \geqslant \frac{\epsilon}{2}$ and $r(-t) = r(t) - t$. The local Seidel-Dehn twist is defined as

$$\tau(u,v) = \begin{cases} \sigma_{2\pi r'(||u||)}(u,v), u \neq 0 \\ (0,-v), u = 0 \end{cases}$$

For a closed symplectic 4-manifold (M,ω) and a Lagrangian sphere $L \subset M$, via the Weinstein neighborhood theorem, we get a symplectomorphism τ_L of (M,ω) by implanting the compactly supported τ. τ_L is called a Seidel-Dehn twist whose symplectic isotopy class only depends on the Lagrangian isotopy class of L. A construction of canonical lift for squared Seidel-Dehn twist in $\pi_1(\mathcal{S}_\omega)$ is presented in [33].

By [Corollary 2.9, [34]], the squared Seidel-Dehn twists are (potentially) fragile symplectic mapping classes in the following sense.

Definition 2.3 ([34])**.** Suppose ϕ is a symplectomorphism of (M,ω). ϕ is said to be potentially fragile if

- There is a smooth path ω^s of symplectic forms for $s \in [0, c_0)$ such that $\omega^0 = \omega$.
- There is a path ϕ^s of diffeomorphisms such that $(\phi^s)^*\omega^s = \omega^s$, $\phi^0 = \phi$.
- For $s > 0, \phi^s$ is symplectically isotopic to the identity.

ϕ is called fragile if ϕ is not isotopic to the identity in $\mathrm{Symp}(M,\omega)$.

Lemma 2.4. *Let ϕ be a potentially fragile symplectomorphism of (M,ω) and let ℓ be a loop in $\mathcal{S}_\omega$ based at ω such that its monodromy defined in* (1.2) *is isotopic to ϕ. There is a loop of diffeomorphisms $\{f_t | t \in S^1\}$ based at id_M such that*

- *There is a disk family D of symplectic forms in $\mathcal{S}(M)$ such that $\partial D = \{(f_t^{-1})^*\ell_t | t \in S^1\}$ defines the symplectic mapping class of ϕ.*
- *The symplectic classes on D are arbitrarily close to $[\omega]$.*

Proof. Since $\phi^s \in \mathrm{Symp}(M,\omega^s)$ for all $s \in [0,c_0)$, we can construct a smooth cylinder family of symplectic forms $\{\omega_t^s | t \in S^1, s \in [0,c]\}$ for some $c < c_0$ such that $\{\omega_t^s\} \subseteq \mathcal{S}_{\omega^s}$ and its monodromy is isotopic to $[\phi^s] \in \pi_0(\mathrm{Symp}(M,\omega^s))$.

The potential fragility of ϕ implies that ϕ^c is isotopic to identity in $\mathrm{Symp}(M,\omega_c)$, so we can take a loop of diffeomorphisms f_t based at $f_0 = id$ such that the loop $\{\omega_t^c | t \in S^1\}$ is homotopic to the loop $\{f_t^*\omega^c | t \in S^1\}$ in $\mathcal{S}_{\omega^c}$. Now consider the cylinder family of symplectic forms

$$\{(f_t^{-1})^*\omega_t^s | t \in S^1, s \in [0,c]\}.$$

The loop $\{(f_t^{-1})^*\omega_t^c | t \in S^1\}$ is homotopic to the constant loop at ω^c and hence bounds a disk in $\mathcal{S}_{\omega^c}$. We extend the cylinder family to a disk family $\mathcal{D}$ of symplectic forms in $\bigcup_{s\in[0,c]} S_{\omega^s}$. Taking c small, the symplectic classes on the disk family can be assumed to be arbitrarily close to $[\omega^0] = [\omega]$. □

Question 2.5. Let ω be a Kähler type form. a) If there is an $\omega-$Lagrangian L, is there a spherical Hodge type twist in the class $\epsilon = [L]$?

b) If there is a there a spherical Hodge type twist in the class ϵ, is there a $\omega-$Lagrangian sphere in the class ϵ?

We have positive answers for 2.5(a) for Kähler type forms on X. Let S be a projective $K3$ surface with an A_1-singularity, let X be its minimal resolution, and let C be the exceptional divisor. By [§ 7.5.4, [20]], S admits a smoothing $f : V \to \mathbb{C}$ of the A_1-singularity, in other word, $f^{-1}(0) = S$ and $f^{-1}(z)$ is smooth for each $z \neq 0$. Applying Atiyah's "small resolution" trick, there is a holomorphic family $\{X_z | z \in \mathbb{C}\}$ of K3 surfaces

such that $X_0 \cong X$ and $X_z \cong f^{-1}(z^2)$ [2][6][34][16]. By the property of the Kähler cone of K3 surfaces, we can equip suitable Kähler forms on $\{X_z | z \in \mathbb{D}^2 \subset \mathbb{C}\}$ such that to get a spherical Hodge type twist.

In general, it is unknown for Kähler type forms on $\tilde{X}$.

2.2. **Elliptic Hodge type twists and elliptic twists.** In this section, we study elliptic twists introduced in [37]. Elliptic twists are constructed in [37] for the nontrivial S^2-bundle over T^2 and shown to be trivial. On the other hand, elliptic twists are also constructed in [37] and [39] for the one point blowup of nontrivial S^2-bundle over T^2 and one point blowup of tori T^4 and shown to be nontrivial. Also, we will study symplectomorphisms representing such elliptic twists in [13].

Let (Z, θ) be a symplectic 4-manifold and C be a symplectic embedded -1 torus in Z. The elliptic divisorial locus $\mathcal{D}_{[C]}$ is defined in [37] as the space of almost complex structures in $\mathcal{A}_\theta$ for which there is a smooth pseudo-holomorphic curve in the class $[C]$. Let J_0 be an almost complex structure compatible to θ such that C becomes a J_0-holomorphic subvariety, $\mathcal{D}_{[C]}$ is nonempty and a submanifold of $\mathcal{A}_\omega$ with real codimension 2 near J_0 [21]. Let $D \subset \mathcal{A}_\theta$ be an embedded disk intersecting $\mathcal{D}_{[C]}$ once transversally at J_0 and let $\ell : S^1 \cong \partial D \to \mathcal{A}_\theta$ be the path of almost complex structures defined by ∂D. Suppose $l(t)$ is tamed by $\{\omega_t | t \in S^1\}$ and $\langle [\omega_t], [C] \rangle \leqslant 0$. By Moser's isotopy, there is a path $\{f_t | t \in [0, 2\pi]\}$ of diffeomorphisms such that $f_t^* \omega_{e^{it}} = \omega_1, t \in [0, 2\pi]$.

Definition 2.6. The diffeomorphism $f_{2\pi}$ is a symplectomorphism of (Z, ω_1), we call it an elliptic twist and denote T_C by its symplectic mapping class in $K(Z, \omega_1)$.

Clearly, the elliptic twist construction is not as rigid as the (squared) Seidel-Dehn twist, even the class of the symplectic form ω_1 is not specified by C. Note however that ω_1 is deformation equivalent to θ since both θ and ω_1 are tamed by some elements in $\mathcal{A}_\theta$.

Lemma 2.7. *Suppose that $\langle \omega_1, [C] \rangle < 0$, then there is a small neighborhood $U \subset \mathcal{S}(M)$ of ω_1 in such that for each $\omega' \in U$, there is an elliptic twist $f \in K(M, \omega')$.*

Proof. Let $\mathcal{K}^t(M, J) = \{[\omega] \in H^2(M; \mathbb{R}) | J \text{ is tamed by } \omega\}$ be the tamed cone of J. Since ℓ is compact subset of $\mathcal{A}_{\omega_1}$ and $[\omega_1] \in \mathcal{K}^t(M, \ell(t))$ for each $t \in S^1$, there is a small neighborhood $U \subset \mathcal{S}(M)$ of ω_1 such that $U \subset \mathcal{K}^t(M, \ell(t))$ for each t. By shrinking U if necessary, we assume that $\forall \omega' \in U, \langle \omega', [C] \rangle \leqslant 0$.

For each $t \in S^1$, the space of cohomologous symplectic forms which tame $\ell(t)$ is convex. Then for each element $\omega' \in U$, we can take a smooth symplectic loop $\{\omega'_t | t \in S^1\}$ such that

(1) ω'_t tames $\ell(t)$ for each $t \in S^1$,
(2) $[\omega'_t] = [\omega']$,

which defines an elliptic twist. □

In this paper, we focus on a special case of elliptic Hodge type twist.

Lemma 2.8. *Let (M, ω) and $D = \{(\omega_z, J_z)\}$ be as in Definition 2.1, we suppose that for any $z \in \mathbb{D}^2 \backslash \{0\}$, $[\omega]$ is contained in the Kähler cone of (M, J_z). The corresponding elliptic Hodge type twist is an elliptic twist.*

Proof. Set $D' = \{z \in \mathbb{D}^2 || z| \leqslant r\}$ for some small $r > 0$, we can assume that ω_0 tames J_z for each $z \in D'$, then $J_z \in \mathcal{A}_{\omega_0}$ for each $z \in D'$. Since $\mathrm{PD}(\epsilon)$ is represented by a smooth elliptic curve in (M, J_0) and ϵ is not in the kernel of $\Omega_*(\partial_t)$, $\{J_z | z \in \mathbb{D}'\}$ intersects $\mathcal{D}_{[C]}$ transversally. Let $\{\omega_z | z \in \partial D'\}$ be a loop of Kähler forms such that $[\omega_z] = [\omega]$, then it defines an elliptic twist.

Let $\ell = \{\omega_z | z \in \partial D\}$ be as in Definition 2.1. Since $[\omega]$ is contained in the Kähler cone of J_z for each $z \in D \backslash \mathrm{int}(D')$, we can extend ℓ and $\{\omega_z | z \in \partial D'\}$ to a cylinder family $\{\omega_z | z \in D \backslash \mathrm{int}(D')\}$ of Kähler forms in the class $[\omega]$. Applying parametric Moser isotopy, we can see that ℓ and $\{\omega_z | z \in \partial D'\}$ define isotopic symplectic mapping classes. □

The two lemmas illustrate the persistence feature of elliptic (Hodge type) twists when $[\omega]$ varies.

Finally, we also mention the connection with the ball swapping construction in the non-minimal case. In [44], the ball swapping class is defined for non-minimal symplectic manifolds. Given two symplectic ball embeddings $\iota_0, \iota_1 \colon \coprod_{i=1}^n B(r_i) \longrightarrow M$ such ι_0 is isotopic to ι_1 through a Hamiltonian path $\{\iota_t | t \in [0,1]\}$ and the image of ι_0 coincides with the image of ι_1. Set

$$M^{\#\iota_j} = \left(M \backslash \iota_j \left(\coprod_{i=1}^n \mathrm{Int}\, B_i \right) \right) / \sim, \quad i = 0, 1,$$

where $B_i = B(r_i)$ is a standard 4-ball of size r_i, and $\sim$ collapses the orbits of the natural S^1-action on $\partial B_i = S^3$. Then the Hamiltonian isotopy $\{\iota_t | t \in [0,1]\}$ induces a symplectomorphism $\tilde{\tau}_\iota$ of $\tilde{M} := M^{\#\iota_0} = M^{\#\iota_1}$. We call τ_ι a *ball-swapping symplectomorphism* on $\tilde{M}$, and we denote a *ball-swapping class* by the symplectic mapping class defined by τ_ι.

Let (M, ω) be a non-minimal Kähler type 4-manifold, and let $\epsilon \in H^2(M; \mathbb{Z})$, $\ell = \partial D \subset \mathcal{S}_\omega$, $D = \{(\omega_z, J_z) | z \in \mathbb{D}^2\} \subset \mathcal{S}(M)$ be as in Definition 2.1. Since $b^+(M) > 1$, (M, ω) admits unique minimal model $(\bar{M}, \bar{\omega})$, and there is a disk family $\{(\bar{\omega}_z, \bar{J}_z) | z \in \mathbb{D}^2\}$ of Kähler structures on $\bar{M}$. If the cohomology class of $\bar{\omega}_z$ is independent of z, then the ϵ-Hodge twist of (M, ω) is a ball-swapping class.

3. Spherical and elliptic Hodge type twists on $(\tilde{X}, \tilde{\omega})$

3.1. **Kähler geometry of (X, j) and the period domain Ω.** Let (X, j) be a complex K3 surface. By [38], (X, j) is a Kähler surface and hence we can pick a Kähler class κ_0 of (X, j). Let $\mathcal{C}_{X,j} \subset H^{1,1}(X, j)$ be one of the connected components of the classes in $H^{1,1}(X, j)$ with positive square that contains the Kähler class κ_0. Inside $H^{1,1}(X, j)$, there are also the Neron-Severi group $NS(X, j) = H^{1,1}(X, j) \cap H^2(X; \mathbb{Z})$ and $\Delta(X, j) = \{l \in NS(X, j) | l^2 = -2\}$. Let

$$\Delta^+(X, j, \kappa_0) = \{d \in \Delta(X, j) | \langle \kappa_0, d \rangle > 0\}$$

and set $\Delta^+(X, j, \kappa_0)$ similarly. Next, we describe the effective classes.

Proposition 3.1. *[Propositions* 3.7 *and* 3.8 *in § VIII, [5]]. Let (X, j) be a complex K3 surface.*

(1) If $d \neq 0 \in NS(X, j)$ and $\langle d, d \rangle \geqslant -2$, then d or $-d$ is effective.

(2) *If d is irreducible, then $d \neq 0$, and $\langle d, d\rangle \geqslant -2$ and equality holds precisely when $d \in \Delta^+(X, j, \kappa_0)$,*
(3) *The set of effective classes is the semigroup generated by $\Delta^+(X, j, \kappa_0)$ and the integral points in the closure of $\mathcal{C}_{X,j}$.*

Since κ_0 pairs positively on any effective class, it follows from Part (1) of Proposition 3.1 that $\langle \kappa, d\rangle \neq 0$ for any $d \in \Delta(X, j)$. Therefore, we have the decomposition

$$\Delta(X, j) = \Delta^+(X, j, \kappa_0) \coprod \coprod \Delta^-(X, j, \kappa_0).$$

Note the subset $\Delta^+(X, j, \kappa_0)$ is exactly the set of nodal classes, therefore the decomposition is independent of the choice of the Kähler class κ_0. We will simply denote $\Delta^+(X, j, \kappa_0)$ by $\Delta^+(X, j)$. The Kähler cone is, in turn, completely determined by $\Delta^+(X, j)$ [Corollary 3.9 in § VIII, [5]].

Proposition 3.2. *The Kähler cone of (X, j) is the following subcone of $\mathcal{C}_{X,j}$,*

$$\mathcal{C}^+_{X,j} = \{y \in \mathcal{C}_{X,j} | \langle y, d\rangle > 0 \text{ for all } d \in \Delta^+(X, j)\}.$$

3.1.1. *The period domain Ω.* Fix the even unimodular lattice

$$(\Lambda, \langle , \rangle) = -E_8 \oplus -E_8 \oplus H \oplus H \oplus H$$

of signature $(3, 19)$. Set $\Lambda_{\mathbb{R}} = \Lambda \otimes \mathbb{R}$ and $\Lambda_{\mathbb{C}} = \Lambda \otimes \mathbb{C}$.

For a complex K3 surface (X, j), $H^2(X; \mathbb{Z})$ is isometric to Λ as a lattice. A choice of such an isometry $\alpha : H^2(X; \mathbb{Z}) \to \Lambda$ is called a *marking* of (X, j). The isometries $H^2(X, \mathbb{R}) \to \Lambda_{\mathbb{R}}$ and $H^2(X, \mathbb{C}) \to \Lambda_{\mathbb{C}}$ induces by α will also be written as α.

Definition 3.3. The period domain for marked complex K3 surfaces is

$$\Omega = \{u \in \mathbb{P}(\Lambda_{\mathbb{C}}) | \langle u, u\rangle = 0, \langle u, \bar{u}\rangle > 0\}.$$

For any $a \neq 0 \in \Lambda_{\mathbb{R}}$, let $Z_a = \{u \in \mathbb{P}(\Lambda_{\mathbb{C}}) | \langle a, u\rangle = 0\}$ be the corresponding hyperplane.

The period domain Ω is an open subset of the non-singular quadric $\{u \in \mathbb{CP}^{21} | \langle u, u\rangle = 0\}$, hence a complex manifold of dimension 20. Note that a point $u \in \Omega$ determines a Hodge structure on $\Lambda_{\mathbb{C}}$:

$$H^{2,0} = \mathbb{C}u, \quad H^{0,2} = \mathbb{C}\bar{u}, \quad H^{1,1} = (H^{2,0} \oplus H^{0,2})^{\perp}.$$

The period map pr associates to a marked complex K3 surface a point $\phi_{(X,j)}$ in Ω, where $\phi_{(X,j)}$ is a generator of $H^{2,0}(X, j)$. It can be considered as a surjective map from the space of marked complex K3 surfaces to the period domain. Hence Ω is a "universal parameter space" of all complex structures, although different points in the period domain may represent isomorphic complex structures. For $u \in \Omega$ we denote the complex K3 surface by X_u.

Lemma 3.4. *For a marked K3 surface (X, j, α) parametrized by $u \in \Omega$, a class $x \in H^2(X; \mathbb{Z})$ is in $H^{1,1}(X, j)$ if and only if $\langle \alpha(x), u\rangle = 0$.*

We are interested in two families of classes $x \in H^2(X; \mathbb{Z}) \cap H^{1,1}(X, j)$: $x^2 = -2$, and primitive x with $x^2 = 0$. By Proposition 3.1, x or $-x$ is effective.

Since $\Delta(X, j)$ is determined by numerical conditions, we can write it as $\Delta(X, j) = \Delta(u)$.

3.2. **The geometry of** (X, κ)**.** We fix a class $\kappa \in \Lambda_{\mathbb{R}}$ with $\kappa^2 > 0$. Following [40] and [39], we introduce various period domains adapted to κ. First, we introduce hyperplanes in Ω.

3.2.1. *The κ-polarized period domain $\mathcal{M}_\kappa^+$.*

Definition 3.5. Let $\overline{\mathcal{M}}_\kappa = \Omega \cap Z_\kappa$, namely, $\overline{\mathcal{M}}_\kappa = \{u \in \Omega | \langle u, \kappa \rangle = 0\}$.

Lemma 3.6. *[40] $\overline{\mathcal{M}}_\kappa$ is a complex manifold of dimension* 19*, consisting of two contractible components ([5],[40]), which are exchanged by the map $u \to \bar{u}$. One of them has the property that for some u in this component, $\kappa \in H^2(X_u; \mathbb{R})$ is a Kähler class. We take this component and denote it by $\overline{\mathcal{M}}_\kappa^+$.*

Proof. The argument to show that $\overline{\mathcal{M}}_\kappa^+$ is contractible was communicated to us by Smirnov. $\overline{\mathcal{M}}_\kappa^+$ can be identified with (a connected component of) the space of positive-definite 2-dimensional spaces in the real space of signature $(2, 19)$. It can be identified with a star-shaped set. □

Definition 3.7. For each $a \neq 0 \in \Lambda_{\mathbb{R}}$, let H_a be the divisor $Z_a \cap \overline{\mathcal{M}}_\kappa^+$ of $\overline{\mathcal{M}}_\kappa^+$. The κ-polarized period domain is $\mathcal{M}_\kappa^+ = \overline{\mathcal{M}}_\kappa^+ - \cup_{\gamma \in S(X,\kappa)} H_\gamma$.

Note that by Lemma 3.4, $\gamma \in S(X, \kappa)$ is in $\Delta(u)$ if and only if $u \in H_\gamma$. Therefore, $u \in \mathcal{M}_\kappa^+$ represents a K3 surface with κ a Kähler class by Proposition 3.2. Moreover, by [40] there is a smooth universal family

$$\mathcal{X} \to \mathcal{M}_\kappa^+$$

of κ-polarized surfaces.

Since the set of divisors $\{H_\gamma | \gamma \in S(X, \kappa)\}$ is local finite, we have

Lemma 3.8 (Lemma 5, [40])**.** *$\mathcal{M}_\kappa^+$ is an open submanifold of $\overline{\mathcal{M}}_\kappa^+$.*

3.2.2. *The subspace $\mathcal{M}_\kappa^{gen}$ of generic complex structures.* Recall $\mathcal{M}_\kappa^{gen}$ is introduced in Definition 1.3 by the Picard number condition: $NS(X_u) = H^{1,1}(X_u) \cap H^2(X, \mathbb{Z}) = \mathbb{R}\kappa \cap H^2(X, \mathbb{Z})$.

Lemma 3.9. *$\mathcal{M}_\kappa^{gen}$ is path connected and dense in $\mathcal{M}_\kappa^+$.*

Proof. When κ is a multiple of some primitive integral class a, we can see for $u \in \mathcal{M}_\kappa^{gen}$, $Pic(X_u) = \mathbb{Z}[a]$, so its Picard number is 1. And when κ is not a multiple of any integral class, Picard number is zero. In both cases, $\mathcal{M}_\kappa^{gen}$ is the complement of the union of countably many hyperplane sections in $\mathcal{M}_\kappa^+$ so it is dense in $\mathcal{M}_\kappa^+$.

To show that $\mathcal{M}_\kappa^{gen}$ is path connected, we just need to prove the path connectedness in a local chart $\mathbb{C}^{19}$ of $\overline{\mathcal{M}}_\kappa^+$, and in $\mathbb{C}^{19}$, we let hyperplanes of $\mathcal{M}_\kappa^+$ be $H_1, H_2, ...$ Let x, y be points in $\mathbb{C}^{19} \backslash \cup H_i$, and $I_0 : [0, 1] \to \mathbb{C}^{19}$ be a path connecting x and y. We can find a small number $\epsilon > 0$ such that there is a path $I_1; [0, 1] \to \mathbb{C}^{19}$ such that

- $I_1(0) = x, I_1(1) = y$.
- $|I_0(t) - I_1(t)| < \epsilon, \forall t \in [0, 1]$.
- distance between I_1 and H_1 is bigger than $\frac{\epsilon}{2}$.

Similarly, we can find an interval I_2 such that $|I_1 - I_2| < \frac{\epsilon}{4}$ and distance $d(I_2, H_2) > \frac{\epsilon}{8}$. We can continue the procedure and $\{I_k | k \in \mathbb{Z}_+\}$ will converge to a path which is disjoint from any H_i. □

3.2.3. *The subdomain $\mathcal{M}^+_{\kappa,\lambda} \subset \mathcal{M}^+_\kappa$.*

Definition 3.10. Fix a real positive λ, set

$$\mathcal{M}^+_{\kappa,\lambda} := \{u \in \mathcal{M}^+_\kappa \mid \kappa - \lambda e \in \mathcal{C}^+_{\tilde{X}_{u,p}} \text{ for any } p \in X_u\}$$

where e is the class of the exceptional divisor of the blowup at p.

A pair in $\mathcal{M}^+_{\kappa,\lambda} \times X$ parametrizes a complex structure of $\tilde{X}$ with a Kähler form in $\kappa - \lambda e$.

Lemma 3.11. *$\mathcal{M}^+_{\kappa,\lambda}$ is an open subset of $\mathcal{M}^+_\kappa$.*

Proof. Let $\mathcal{T}^+_{\tilde{X}_{u,p}}$ be the tamed cone of $\tilde{X}_{u,p}$. By [Corollary 1.1,[31]], we have

$$\mathcal{C}^+_{\tilde{X}_{u,p}} = \mathcal{T}^+_{\tilde{X}_{u,p}} \cap H^{1,1}(\tilde{X}_u; \mathbb{R}).$$

Since $\kappa - \lambda e \in H^{1,1}(\tilde{X}_{u,p}; \mathbb{R})$ for each $u \in \mathcal{M}^+_\kappa$, $\kappa - \lambda e \in \mathcal{T}^+_{\tilde{X}_{u,p}}$ if and only if $\kappa - \lambda e \in \mathcal{C}^+_{\tilde{X}_{u,p}}$.

Since the taming condition is open and the parameter space X is compact, we can see that $\mathcal{M}^+_{\kappa,\lambda}$ is an open subset of $\mathcal{M}^+_\kappa$. □

3.2.4. *Kahler type ball embeddings and* $\mathrm{Se}(\kappa)$. Let X_u be a complex K3 surface and let x be a point of it, we blowup X_u and get $\tilde{X}_{u,x}$. Since the blowup of a Kähler surface is Kähler, there exists some λ_u such that $\kappa - \lambda_u e$ is a Kähler class of $\tilde{X}_{u,Z}$ if λ_u is controlled by $\sqrt{\kappa^2}$ and the Seshadri constant.

Let ω be a Kähler form on X, let ω_0 be a Kähler form on the standard ball B^4 of size r. Let $Emb^r_x(X, \omega)$ be the space of $\iota : B^4 \hookrightarrow X_u$ is a holomorphic ball embedding centered at x such that $\iota^*\omega = \omega_0$.

Definition 3.12. The κ-Seshadri constant of X is the function $\mathrm{Se} : V^+ \to \mathbb{R}^{>0}$ as

$$\mathrm{Se}(\kappa) := \pi \inf_{u \in \mathcal{M}^{gen}_\kappa} \min_{x \in X_u} \sup_{Emb^r_x(X,\omega) \neq \varnothing, [\omega] = \kappa} r^2 \tag{3.1}$$

Now we introduce some basic properties of the κ-Seshadri constant.

Lemma 3.13. *For each $x \in X_u$ and $\lambda < \mathrm{Se}(\kappa)$, $\kappa - \lambda e$ is a Käler class of $\tilde{X}_{u,x}$.*

- *Suppose $\lambda \in \big(0, \mathrm{Se}(\kappa)\big)$. We have $\mathcal{M}^+_{\kappa,\lambda} \supset \mathcal{M}^{gen}_\kappa$ and $\mathcal{M}^+_{\kappa,\lambda}$ is an open, dense and path connected subspace of $\mathcal{M}^+_\kappa$.*
- *$\mathrm{Se}(\kappa) = \sqrt{\kappa^2}$ when κ is not a multiple of any rational class.*

Proof. By [Proposition 5.16, [14]], $Emb^r_x(X_u, \omega_u) \neq \varnothing$ implies that there is a Kähler form $\tilde{\omega}_u$ on $\tilde{X}_{u,x}$ such that $[\tilde{\omega}_u] = \kappa + \pi r^2 e = \kappa - \lambda e$. By Definition 3.12, such Kähler forms exist for any $u \in \mathcal{M}^{gen}_\kappa$ and any $x \in X_u$, so $\mathcal{M}^+_{\kappa,\lambda} \supset \mathcal{M}^{gen}_\kappa$ when $\lambda \in \big(0, \mathrm{Se}(\kappa)\big)$.

Since $\mathcal{M}^{gen}_\kappa \subset \mathcal{M}^+_{\kappa,\lambda}$, $\mathcal{M}^+_{\kappa,\lambda}$ must be dense in $\mathcal{M}^+_\kappa$. Since $\mathcal{M}^{gen}_\kappa$ is path connected and $\mathcal{M}^+_{\kappa,\lambda}$ is open and hence local path connected, $\mathcal{M}^+_{\kappa,\lambda}$ is also path connected.

We can get the second bullet from [Theorem 3.22, [14]]. □

Let S be a smooth projective variety and L an ample divisor. For $\mathcal{C}_x$ the set of irreducible curves through x, the Seshadri constant is defined as

$$\epsilon(S,x,L) := \inf_{C\in\mathcal{C}_x} \frac{L\cdot C}{\mathrm{mult}_x C}, \qquad \epsilon(S,L) := \inf_{x\in S}\epsilon(x,L).$$

A K3 surface X_u is projective if $L \in NS(X_u)$, if $u \in \mathcal{M}_\kappa^{gen}$ and $\kappa = r\cdot L$ for some real number r, then $NS(X_u) = \mathbb{Z}L$.

Theorem 3.14. *([19][22]) Let (X,j) be a smooth projective K3 surface with $Pic(X,j) = \mathbb{Z}[L]$. Let $\alpha = \lfloor\sqrt{L^2}\rfloor$ be the lower floor of $\sqrt{L^2}$. Denote $\epsilon((X,j),L)$ by $\epsilon(j,L)$. Then either $\sqrt{L^2} \geqslant \epsilon(j,L) \geqslant \alpha$, or $\epsilon(j,L) < \alpha$. When $\epsilon(j,L) < \alpha$, there are two cases: $\epsilon(j,L) = \alpha - \frac{2}{\alpha+1}$ if $L^2 = \alpha^2+\alpha-2$; or $\epsilon(j,L) = \alpha - \frac{1}{2\alpha+1}$ if $L^2 = \alpha^2 + \frac{\alpha-1}{2}$.*

Lemma 3.15. *If L is a primitive integral class, then $\mathrm{Se}(L)$ will be equal to*

$$\inf_{u\in\mathcal{M}_\kappa^{gen}}\{\epsilon(X_u,L)|NS(X_u) = \mathbb{Z}[L], L \text{ is a Kähler class of } X_u\}.$$

If $\kappa = rL$ for a primitive integral class L and a positive real number r, then $\mathrm{Se}(\kappa) = r\mathrm{Se}(L)$.

For a primitive integral class κ, we have $\mathrm{Se}(\kappa) \geqslant \frac{\sqrt{2}}{2}\sqrt{\kappa^2} \geqslant 1$. Both equalities hold if and only if $\kappa^2 = 2$.

Proof. By [Proposition 5.13, Proposition 5.15 and Proposition 5.16, [14]], $Emb_x^r(X,\omega) \neq \varnothing$ if and only if $\kappa - \pi r^2 e$ is a Kähler class of $\tilde{X}_{u,x}$. By [Theorem 0.1, [10]], $\kappa - \lambda e$ is a Kähler class of $\mathcal{M}_\kappa^{gen}$ if and only if

$$\langle \kappa - \lambda e, C - mult_x e\rangle > 0, \quad C \in \mathcal{C}_x.$$

By the definition of Seshadri constant,

$$\epsilon(X_u,x,L) = \pi \min_{x\in X_u} \sup_{Emb_x^r(X,\omega)\neq\varnothing,[\omega]=\kappa} r^2.$$

And by the definition of $\mathrm{Se}(\kappa)$, it's easy to see that $\mathrm{Se}(rL) = r\mathrm{Se}(L)$.

Let $L = \kappa$, we first observe that we always have $\alpha = \lfloor\sqrt{L^2}\rfloor \geqslant \sqrt{\frac{L^2}{2}}$, and equality occurs if and only if $L^2 = 2$. So we just need to consider the two exceptional cases in Theorem 3.14. Assume first that $\big(\epsilon(j,L),L^2\big) = (\alpha - \frac{2}{\alpha+1}, \alpha^2+\alpha-2)$. If $\alpha = 1$, then $L^2 = 0$, which is impossible. If $\alpha = 2$, then $L^2 = 4$ and $\sqrt{\frac{L^2}{2}} = \sqrt{2}$, while $\epsilon(j,L) \geqslant 2$ by [22][19]. When $\alpha \geqslant 3$,

$$\epsilon(j,L)^2 - \frac{L^2}{2} = \alpha^2 - \frac{4\alpha}{\alpha+1} + \frac{4}{(\alpha+1)^2} - \frac{1}{2}(\alpha^2+\alpha-2) > \frac{\alpha^2}{2} - 4 - \frac{\alpha}{2} + 1 \geqslant 0.$$

Now assume $\big(\epsilon(j,L),L^2\big) = (\alpha - \frac{1}{2\alpha+1}, \alpha^2 + \frac{\alpha-1}{2})$. If $\alpha = 1$ and 2 then $L^2 = 1$ and $5/2$ respectively, which is impossible. When $\alpha \geqslant 3$, we have

$$\epsilon(j,L)^2 - \frac{L^2}{2} \geqslant \alpha^2 - \frac{2\alpha}{2\alpha+1} - \frac{\alpha^2}{2} - \frac{\alpha-1}{4} > \frac{\alpha^2}{2} - \frac{\alpha}{4} - 1 > 0.$$

□

Remark 1. The Seshadri constant $\epsilon(L, X_u)$ can be smaller than the lower bounds in Theorem 3.14 when its Picard number is greater than 1. For specific examples, see [Proposition 1.1 and Theorem 1.2, [19]].

Proof of Lemma 1.4. The first two bullets are [Theorem 5.16 [14]] and [Theorem 3.22, [14]]. We get the third and fourth bullets from Lemma 3.13 and Lemma 3.15. □

3.3. The index sets $S(X, \kappa)$ and $T_\lambda(X, \kappa)$.

3.3.1. *Lagrangian spheres and the index set $S(X, \kappa)$.* For a smooth 4-manifold Y, let $S(Y)$ be the set of square -2 classes represented by smooth spheres. For a K3 surface X, a well-known fact is $S(X) = \{\gamma \in H^2(X; \mathbb{Z}) | \gamma^2 = -2\}$. By the property of Λ and diffeomorphism group of K3 surface [11], we have

Lemma 3.16. $\mathrm{Diff}(X)$ *acts transitively on integral classes with the same square and divisibility. Thus* $\mathrm{Diff}(X)$ *acts transitively on* $S(X)$.

Lemma 3.17. *If κ is a multiple of some integral class, then $S(X, \kappa)$ is infinite.*

Proof. Let $\kappa \in ((-E_8)^{\oplus 3} \oplus H^{\oplus 2}) \otimes \mathbb{R}$ be a positive class, and suppose that there is a decomposition

$$((-E_8)^{\oplus 3} \oplus H^{\oplus 2}) \otimes \mathbb{R} = W \oplus W^\perp$$

such that $\kappa \in W$ and $W^\perp$ contains a subspace V isomorphic to $(-E_8 \oplus H) \otimes \mathbb{R}$. Since there are infinitely many roots in $-E_8 \oplus H$, and the set of roots in V is a subset of $S(X, \kappa)$, $S(X, \kappa)$ is infinite in this case.

By Lemma 3.16, κ is equivalent to $p\mathcal{R}_1 + qT_1 \in (-E_8)^{\oplus 2} \oplus H^{\oplus 3}$, so we have the decomposition stated above, and $S(X, \kappa)$ is infinite. □

Lemma 3.18. *Let X be the K3 surface. Under the natural inclusion $H_2(X; \mathbb{Z}) \subset H_2(\tilde{X}; \mathbb{Z})$, we have $S(\tilde{X}) = S(X)$. Consequently, we have $S(\tilde{X}, \kappa - \lambda e) = S(X, \kappa)$.*

Proof. If $(A - be)^2 = A^2 - b^2 = -2$, then b is an even number since A^2 is even. Since $K(\tilde{X}) = e$, we have $\langle K(\tilde{X}), A - be \rangle = b$. We know that the SW basic classes of $\tilde{X}$ are $\pm e$. By the SW adjunction inequality of Ozsváth–Szabó for 4-manifolds with $b^+ > 1$ and classes with negative squares, if $A - be$ is represented by a smooth surface, then its genus will satisfy the following:

$$2g(A - be) - 2 \geqslant |K(\tilde{X}) \cdot (A - be)| + (A - be)^2 = |b| + A^2 - b^2 = |b| - 2.$$

If $b \neq 0$, then $|b| - 2 \geqslant 0$ and $g(A - be) > 0$. □

Remark 2. For a Kähler type form ω, every class in $S(\tilde{X}, [\omega] - \lambda e) = S(X, [\omega])$ is represented by a $\tilde{\omega}$-Lagrangian sphere L in $(\tilde{X}, \tilde{\omega})$ for small λ. Moreover, by [Theorem 5.4, [35]], τ_L^2 is nontrivial in $K(\tilde{X}, \tilde{\omega})$.

Question 3.19. Are there ϵ-Lagrangian spheres on $(\tilde{X}, \tilde{\omega})$ whenever $\epsilon \in S(X, \kappa)$? Or when there is a spherical Hodge type twist?

3.3.2. *Symplectic tori and the index set $T_\lambda(X,\kappa)$.*

Definition 3.20. Let $T(X)=\{\delta\in H^2(X;\mathbb{Z})|\delta^2=0 \text{ and } \delta \text{ is primitive}\}$, and let $T(X,\kappa)$ be $\{\delta\in T(X)|\langle\kappa,\delta\rangle>0\}$. We omit X when there is no ambiguity.

Lemma 3.21. *Every class δ in $T(X)$ is represented by a smooth torus. For $\delta\in T(X,\kappa)$ and $u\in H_\delta$, the class δ is effective on X_u. For generic $u\in H_\delta$, the class δ is also nef and hence there is an elliptic fibration on X_u with fibers in the class δ.*

Proof. As in [12], let $\{D_{i,1};D_{i,2};\mathcal{R}_1,T_1;\mathcal{R}_2,T_2;\tilde{F},W|i=1,2,...,8\}$ be a system of generators of $(-E_8)^{\oplus 2}\oplus H^{\oplus 3}$.

Given $\delta\in T(X)$ it follows from Lemma 3.16 that we can assume that $\delta=\mathcal{R}_1$, which is represented by a smooth torus. By Lemma 3.4 for u in H_δ, there exists a κ-polarized K3 surface which admits a curve C in class δ. Moreover, by [Lemma 2.1, [20]], the linear system $|C|$ is of dimension 1. By [Corollary 3.6, [20]], we know generic members in the family are smooth elliptic curves.

For a generic $u\in H_\delta$ the class δ is nef and so the linear system $|C|$ has no base locus. □

Definition 3.22. Let τ be the function $\tau: V^+\to\mathbb{R}^{\geqslant 0}$ defined by $\tau(\kappa):=\inf_{\delta\in T(X,\kappa}\langle\kappa,\delta\rangle$.

Notice that $\tau(r\cdot\kappa)=|r|\cdot\tau(\kappa)$ for any real number $r\in\mathbb{R}\backslash\{0\}$.

Lemma 3.23. *Let τ be the function under consideration. Then*

$$\tau(\kappa)=\begin{cases}1, & \text{if } \kappa=L \text{ for some primitive integral class } L,\\ 0, & \text{if } \kappa \text{ is not a real multiple of a rational class.}\end{cases}$$

Proof. Since $\{\langle L,\delta\rangle|\delta\in T(X,L)\}$ consists of positive integers, we have $\tau(L)\geqslant 1$. So it suffices to find a class $\delta_0\in T_L$ with $\langle L,\delta_0\rangle=1$. By Lemma 3.16, we can assume $L=p\mathcal{R}_1+qT_1$, where p,q are coprime positive integers, $p\geqslant q>0$. If we choose $r,s\in\mathbb{Z}$ such that $ps+qr=1$ then $\delta_0=r\mathcal{R}_1+sT_1+\mathcal{R}_2-rsT_2\in T_L$ and $\langle L,\delta_0\rangle=1$.

Let κ be a class which is not a real multiple of a rational class. Let $\{\epsilon_n\}$ be a sequence of positive numbers such that $\epsilon_n<\frac{1}{4n^2}$. As in [Theorem 6.4, [12]], for any $n\in\mathbb{Z}_+$, κ is equivalent to the following form

$$\tilde{\alpha}_0=\sum_{i=0}^{7}k_{i,1}D_{i,1}+\sum_{i=0}^{7}k_{i,2}D_{i,2}+a_1\mathcal{R}_1+b_1T_1+a_2\mathcal{R}_2+b_2T_2+w\tilde{F}+gW\in(-E_8)^{\oplus 2}\oplus H^{\oplus 3}$$

such that the absolute values of $k_{ij},a_i\cdot b_i$ and g are smaller than ϵ_n. Now we let δ be $\tilde{F}+\mathcal{R}_1$ if $|b_1|\leqslant|a_1|$, and let it be $\tilde{F}+T_1$ otherwise. Without loss of generality, we can assume that $\langle\kappa,\delta\rangle>0$. We can see that

$$0<\langle\kappa,\delta\rangle=g+b_1(\text{ or } g+a_1)<\frac{1}{4n^2}+\sqrt{\frac{1}{4n^2}}<\frac{1}{2n}+\frac{1}{2n}=\frac{1}{n}.$$

It follows that $\tau(\kappa)=0$ in this case. □

Recall that $T_\lambda(X,\kappa):=\{\delta\in T(X,\kappa\mid\langle\kappa-\lambda e,\delta-e\rangle=\langle\kappa,\delta\rangle-\lambda<0\}$.

Lemma 3.24. *For any κ which is not a multiple of an integral class L with $L^2 = 2$, $T_\lambda(X,\kappa)$ is nonempty for any $\lambda \in (\tau(\kappa), \sqrt{\kappa^2})$. In particular, when κ is not a real multiple of a rational class, $T_\lambda(X,\kappa)$ is infinite for any $\lambda \in (0, \sqrt{\kappa^2})$.*

Proof. The first statement follows from the definition of $\tau(\kappa)$. The second statement follows from the second statement of Lemma 3.23. □

Remark 3. $\mathrm{Se}(\kappa) \geqslant \tau(\kappa)$ for any positive class κ. And the equality holds if and only if κ is (a multiple of) an integral class of self-intersection 2. When κ is a multiple of $L \in H^2(X;\mathbb{Z})^{prim}$ such that $L^2 = 2$, we can still show that $T_\lambda(X,\kappa) \neq \varnothing$.

Proof of Lemma 1.5. When κ is a multiple of some rational class, $\bar{S}(X,\kappa)$ is infinite by Corollary 3.17. When κ is not a multiple of any rational class, by Lemma 3.24, $T_\lambda(X,\kappa)$ is infinite for any $\lambda < \sqrt{\kappa^2} = \mathrm{Se}(\kappa)$. So $\bar{S}(X,\kappa) \cup T_\lambda(X,\kappa)$ is always infinite. □

3.4. **Constructing Hodge type twists.**

3.4.1. *The subdomain $\mathcal{D}_{\kappa,\lambda}(X) \supset \mathcal{M}^+_{\kappa,\lambda}$.*

Definition 3.25. We define

$$\mathcal{D}_{\kappa,\lambda}(X) := \mathcal{M}^+_\kappa - \bigcup_{\delta \in T_\lambda(X,\kappa)} H_\delta = \overline{\mathcal{M}}^+_\kappa - \bigcup_{\gamma \in S(X,\kappa)} H_\gamma - \bigcup_{\delta \in T_\lambda(X,\kappa)} H_\delta \tag{3.2}$$

Lemma 3.26. *For any λ, we have*

- *$\mathcal{D}_{\kappa,\lambda}(X)$ is open in $\mathcal{M}^+_\kappa$ for any λ.*
- *$H_1(\mathcal{D}_{\kappa,\lambda}(X))$ is generated by meridian loops indexed by classes $\tau \in \overline{S}(X,\kappa) \cup T_\lambda(X,\kappa)$.*
- *$\mathcal{M}^+_{\kappa,\lambda} \subset \mathcal{D}_{\kappa,\lambda}(X)$.*

Proof. The openness of $\mathcal{D}_{\kappa,\lambda}$ follows directly from the local finiteness of the hyperplanes corresponding to both $T_\lambda(X,\kappa)$ and $S(X,\kappa)$, proved in Lemma 4 in [40], Lemma 12 in [39].

The generation statement follows from the contractibility of $\overline{\mathcal{M}}^+_\kappa$ as in [40].

For any $\delta \in T_{\kappa,\lambda}$ consider a blowup surface $\tilde{X}_{u,p}$, where u is a generic point in H_δ and p is a point in X_u. Suppose $\kappa - \lambda' e$ is a Kähler class of $\tilde{X}_{u,p}$, then by Lemma 3.21, p must lie on a curve in the linear system $|\delta|$. Then we have

$$\lambda' < \langle \kappa, \delta \rangle < \lambda,$$

thus $u \notin \mathcal{M}^+_{\kappa,\lambda}$. Since $H_\delta \cap \mathcal{M}^+_{\kappa,\lambda}$ is open in H_δ and the set of generic points in H_δ is not contained in $\mathcal{M}^+_{\kappa,\lambda}$, $H_\delta \cap \mathcal{M}^+_{\kappa,\lambda}$ must be empty, and $\mathcal{M}^+_{\kappa,\lambda} \subseteq \mathcal{D}_{\kappa,\lambda}(X)$. □

To study the relation between $H_1(\mathcal{D}_{\kappa,\lambda}(X))$ and $H_1(\mathcal{M}^+_{\kappa,\lambda})$, we construct the following holomorphic disk families.

Lemma 3.27. *For each $\tau \in ST(X,\kappa,\lambda)$, there is a holomorphic disk D_τ such that*

- *$D_\tau \pitchfork H_\tau$,*
- *$D_\tau \cap H_{\tau'} = \varnothing$ for $\tau' \in ST(X,\kappa,\lambda)\backslash\{\tau\}$,*

- *when κ is not a multiple of an integral class, D_τ is transverse to any $H_a, a \neq 0 \in E$,*
- *when κ is a multiple of a primitive integral class L, D_τ is transverse to any $H_a, a \in E \backslash \mathbb{Z} \cdot L$.*

Proof. We construct local families following Oguiso and [8]. From the local finiteness of hyperplanes $\{H_\tau\}$, we can take a small open subset of $\overline{\mathcal{M}}_\kappa^+$ which is isomorphic to a standard open ball in $\mathbb{C}^{19}$ satisfying the first two conditions. In the local chart of $\overline{\mathcal{M}}_\kappa^+$, let

$$\mathcal{H} = Hol(D, B) \cap \mathcal{C}^0(\bar{D}, B) \tag{3.3}$$

be the space of holomorphic maps from a disk D to an open ball $B \subset \mathbb{C}^{19}$ which is continuous on its closure. $\mathcal{H}$ is an open subset of a complete metric space $Hol(D, \mathbb{C}^{19}) \cap \mathcal{C}^0(\bar{D}, \mathbb{C}^{19})$ with the sup-norm. $\mathcal{H}$ is a Baire space.

By [Lemmas 4.1 and 4.3, [8]], for any $\kappa \notin \mathbb{R} \cdot a$, the hyperplane $H_a = Z_a \cap \overline{\mathcal{M}}_\kappa^+$ is a codimension 1 variety in $\overline{\mathcal{M}}_\kappa^+$ and its singular locus is a smooth submanifold of higher codimension. And the space $\mathcal{H}_a := \{f \in \mathcal{H} | f(D) \text{ is transverse to } H_a\}$ is open and dense in $\mathcal{H}$. As a result, we can consider the disk family as a map to the polarized period domain. Since $H^2(X; \mathbb{Z}) \backslash \{0\}$ is countable, $\bigcap_{a \in H^2(X;\mathbb{Z})} \mathcal{H}_a$ is of second category in $\mathcal{H}$. And disks D_τ with required properties can be obtained by perturbing any disk satisfying the first two conditions. □

Lemma 3.28. *There is a surjection from $H_1(\mathcal{M}_\kappa^{gen})$ to $H_1(\mathcal{D}_{\kappa,\lambda}(X))$. Consequently, the homomorphism*

$$H_1(\mathcal{M}_{\kappa,\lambda}^+) \to H_1(\mathcal{D}_{\kappa,\lambda}(X))$$

is a surjection when $\lambda < \mathrm{Se}(\kappa)$.

Proof. By Lemma 3.26, $H_1(\mathcal{D}_{\kappa,\lambda}(X))$ is generated by links of H_τ where $\tau \in T_{\kappa,\lambda}$ or $S(X, \kappa)$. Let D_τ be a disk in $\overline{\mathcal{M}}_\kappa^+$ intersecting H_τ transversely, as in Lemma 3.27. In this case, $D_\tau \backslash \mathcal{M}_\kappa^{gen}$ is countable. To complete the proof, we just need to find a loop homologous to ∂D_τ which is contained in $\mathcal{M}_\kappa^{gen}$. Since $D_\tau \backslash \mathcal{M}_\kappa^{gen}$ is countable, after possibly shrinking the "radius" of D_τ, we obtain uncountably many pairwise disjoint loops ℓ_t. For some parameter t, the loop ℓ_t lies entirely in $\mathcal{M}_\kappa^{gen}$.

□

Recall $\mathcal{X} \to \mathcal{M}_\kappa^+$, the universal family of κ-polarized surfaces. Following [40], we have the following result based on the Burns-Rapoport Theorem.

Lemma 3.29. *The restriction of $\mathcal{X}$ to each loop in $\mathcal{D}_{\kappa,\lambda}$ is smoothly trivial.*

For each τ, we can introduce the homomorphisms $l_\tau : H_1(\mathcal{D}_{\kappa,\lambda}) \to \mathbb{Z}_2$ as follows. Let γ be a loop in $\mathcal{D}_{\kappa,\lambda}$. Since $\overline{\mathcal{M}}_\kappa^+$ is contractible there is a null-homotopy $f : D \to \overline{\mathcal{M}}_\kappa^+$ of γ. By the local finiteness, $f(D)$ only intersects finitely many varieties H_τ and so we can define

$$l_\tau(\gamma) = \#(f(D) \cap H_\tau) \pmod 2.$$

Let $l_S = \bigoplus_{\gamma \in \bar{S}(X,\kappa)} l_\gamma : H_1(\mathcal{D}_{\kappa,\lambda}) \to \bigoplus_{\gamma \in \bar{S}(X,\kappa)} \mathbb{Z}_2$, and $l_T = \bigoplus_{\delta \in T_{\kappa,\lambda}} l_\delta : H_1(\mathcal{D}_{\kappa,\lambda}) \to \bigoplus_{\delta \in T_{\kappa,\lambda}} \mathbb{Z}_2$.

3.4.2. *Construction of Hodge type twists on* $(\tilde{X}, \tilde{\omega})$. For any (κ, λ) such that $\mathcal{M}_\kappa^{gen} \subset \mathcal{M}_{\kappa,\lambda}^+$, we will construct Hodge type twists on $\tilde{X}$ from the loops in $\mathcal{M}_{\kappa,\lambda}^+$.

Since cohomologous Kähler type forms are diffeomorphic, we can replace $H_n(\mathcal{KS}_{\tilde{\omega}})$ by $H_n(\mathcal{KS}_{\kappa-\lambda e})$ without ambiguity for each $n > 0$.

Lemma 3.30. *We have a homomorphism* $H_1(\mathcal{M}_{\kappa,\lambda}^+) \to H_1(\mathcal{KS}_{\kappa-\lambda e})$, *up to the action of* $\pi_1(\mathrm{Diff}_0)$.

Proof. Take a loop $\{u_t | t \in S^1\}$ in $\mathcal{M}_{\kappa,\lambda}^+$. By Lemma 3.29, we can fix a smooth trivialization of $\mathcal{X}$ over u_t. Take a constant section Z. Then we have a smoothly trivial loop family ℓ of blowup of K3 surfaces with a loop of complex structures $\{J_t\}$ on $X\#\overline{\mathbb{CP}}^2$.

By [Theorem 0.1, [10]], we can take a loop of Kähler forms $\{\tilde{\omega}_t\}$ fiberwise compatible with J_t such that $[\tilde{\omega}_t] = \kappa - \lambda e$. This will define a loop in $\mathcal{KS}_{\kappa-\lambda e}$. And it will not be hard to see that we get a well-defined map in homotopy. □

For any $\tau \in ST(X, \kappa, \lambda)$, let $\epsilon(\tau) \in H^2(\tilde{X}; \mathbb{Z})$ (or ϵ for simplicity) be τ if $\tau \in \bar{S}(X, \kappa)$, or $\tau - e$ if $\tau \in T_\lambda(X, \kappa)$.

Lemma 3.31. *For any* (κ, λ) *such that* $\mathcal{M}_\kappa^{gen} \subset \mathcal{M}_{\kappa,\lambda}^+$ *and* $\tau \in ST(X, \kappa, \lambda)$, *there exists a disk family* $D_{\epsilon(\tau)}$ *of Kähler forms such that*

(1) Each $\ell_\epsilon := \partial D_\epsilon$ *defines an element in* $H_1(\mathcal{KS}_{\kappa-\lambda e})$,
(2) $D_\epsilon \cap H_{\epsilon'} \neq \varnothing$ *if and only if* $\epsilon = \epsilon'$,
(3) D_ϵ *defines a spherical Hodge type twist if* $\epsilon \in \bar{S}(X, \kappa)$, *and* D_ϵ *defines an elliptic Hodge type twist if* $\epsilon \in -e + T_{\kappa,\lambda}$.

Proof. For each $\tau \in ST(X, \kappa, \lambda)$, we can take a holomorphic disk $D_\tau = \{u_b \in \overline{\mathcal{M}}_\kappa^+ | b \in \mathbb{D}^2\}$ such that

(1) $\partial D_\tau \subset \mathcal{M}_\kappa^{gen}$,
(2) The hyperplanes $H_\tau \subset \overline{\mathcal{M}}_\kappa^+$ and disk D_τ satisfy the conditions in Lemma 3.27,
(3) The unique intersection point u_0 of D_τ and H_τ is a generic point in H_τ,

Since the family is a holomorphic submersion, there is a locally holomorphic section Z such that it is constant on ∂D with respect to a smooth trivialization and $Z(u_0)$ is a "good" point with respect to the complex structure in u_0, i.e.

(1) If $\tau \in \bar{S}(X, \kappa)$, then $Z(u_0)$ is not contained in the smooth PD(τ)-curve,
(2) If $\tau \in T_{\kappa,\lambda}$, then $Z(u_0)$ is contained in a smooth fiber of the elliptic fibration in Lemma 3.21.

In other words, there is a smooth PD(ϵ)-curve in $\tilde{X}_{u_0}$.

We let D_ϵ be the blowup family of K3 surfaces along Z. By Lemma 3.30, ∂D_ϵ defines an element in $H_1(\mathcal{KS}_{\kappa-\lambda e})$.

By [40], we can see that ϵ is not in the kernel of $\Omega_*(\partial_t)$. Following Definition 2.1, to construct a Hodge type twist, we just need to take a family of Kähler forms $\{\tilde{\omega}_b | b \in D_\epsilon\}$ such that

(1) $[\tilde{\omega}_t] = \kappa - \lambda e$, $\forall t \in \partial D_\epsilon$,
(2) D_ϵ is $\tilde{\omega}_1$-admissible, i.e. $\langle [\tilde{\omega}_b], \kappa - \lambda e \rangle > 0, \forall b \in D_\epsilon$.

Suppose that $\epsilon \in \bar{S}(X,\kappa)$, there is an effective divisor $\gamma \in S(X,\kappa)$ on $\tilde{X}_{u_0}$ such that γ is a representative of ϵ. By Proposition 3.2, there is a Kähler class $\kappa - c\cdot\gamma$ on X_{u_0} for some small c, and there is a Kähler class $\kappa - c\cdot\gamma - c'e$ on $\tilde{X}_{u_0}$ for some small c'. By the Kodaira-Spencer stability theorem, we can get a smooth family $\{\tilde{\theta}_b, b \in D' \subset D_\epsilon\}$ of Kähler forms on a small disk D' containing 0. Shrinking D' if necessary, we can assume that

- $\forall b, b' \in D'$, the difference between $[\tilde{\theta}_b]$ and $[\tilde{\theta}_{b'}]$ is very small,
- $u_t \in \mathcal{M}_\kappa^{gen}, \forall t \in \partial D'$.

We can define a Kähler family $\{\tilde{\omega}_b | b \in \overline{D_\epsilon \backslash D'} | [\tilde{\omega}_b] = \kappa - \lambda e\}$ such that $\tilde{\omega}_b$ is a Kähler form on $\tilde{X}_{u_b}$. By the convexity of the Kähler cone of a complex manifold, we can construct a cylinder family $A = \{\theta_{s,t} | s \in [0,1], t \in \partial\mathbb{D}' \cong S^1\}$ of Kähler forms such that

- $\theta_{s,t}$ is a Kähler form on $\tilde{X}_t$ for each $t \in \partial\mathbb{D}'$,
- $\theta_{0,t} = \tilde{\theta}_t$,
- $\theta_{1,t} = \tilde{\omega}_t$,
- $[\theta_{s,t}] = s[\tilde{\omega}_t] + (1-s)[\tilde{\theta}_t]$ for any $(s,t) \in [0,1] \times S^1$.

We can "insert" the cylinder family A to glue the Kähler families on D' and $\overline{D_\epsilon \backslash D'}$.

Due to the construction of $\{\theta_{s,t}\}$, to see the disk family is $\tilde{\omega}_1$-admissible, we just need to show that $\langle \kappa - \lambda e, [\tilde{\theta}_b]\rangle > 0$ for any $b \in D'$. Since we assume that D' is small enough, we just need to see that

$$\langle \kappa - \lambda e, \kappa - c\cdot\gamma - c'e\rangle > 0.$$

Since $\langle \kappa, \gamma\rangle = 0$, we have

$$\langle \kappa - \lambda e, \kappa - c\cdot\gamma - c'e\rangle = \kappa^2 - \lambda\cdot c' \geqslant \kappa^2 - \lambda^2 > 0,$$

thus D_ϵ defines a spherical Hodge type twist.

Let $\tau \in T_{\kappa,\lambda}$ and $\epsilon = \epsilon(\tau) = \tau - e$, we'll take a family of Kähler forms $\{\tilde{\omega}_b | b \in D_\epsilon\}$ such that

(1) $[\tilde{\omega}_t] = \kappa - \lambda e$ for $t \in \partial D_\epsilon$.
(2) $\{\tilde{\omega}_b\}$ is $\tilde{\omega}_1$-admissible.

Since $\partial D_\tau \subset \mathcal{M}_\kappa^{gen} \subset \mathcal{M}_{\kappa,\lambda}^+$, by Lemma 3.30, we can take a family of Kähler forms $\{\tilde{\omega}_b\}$ on D_ϵ such that

(1) $[\tilde{\omega}_b] = \kappa - \lambda_b e$ for each $b \in D_\epsilon$,
(2) $\lambda_t = \lambda$ for each $t \in \partial D_\epsilon$,
(3) $\lambda(\tilde{\omega}_b) \leqslant \lambda$ for each $b \in D_\epsilon$.

Since

$$\langle [\tilde{\omega}_b], [\tilde{\omega}_1]\rangle = \langle \kappa - \lambda_b e, \kappa - \lambda e\rangle = \kappa^2 - \lambda\lambda_b \geqslant \kappa^2 - \lambda^2 > 0,$$

D_ϵ defines an elliptic Hodge type twist. □

Lemma 3.32. *U_{TH} contains a connected and dense subset of $\mathcal{KS}(\tilde{X})$.*

Proof. By the construction above, for each $\lambda < \mathrm{Se}(\kappa)$, $\tau \in T_\lambda(X,\kappa)$ and a Kähler type form $\tilde{\omega}$ such that $[\tilde{\omega}] = \kappa - \lambda e$, there is an elliptic Hodge type twist $\partial\ell_{\delta - e}$ on $(\tilde{X}, \tilde{\omega})$. If

κ is not a multiple of any rational class, then $T_\lambda(X,\kappa) \neq \varnothing$ for any $\lambda \in (0, \sqrt{\kappa^2})$, thus $\tilde{\omega} \in U_{TH}$ if $[\tilde{\omega}] = \kappa - \lambda e$. Define

$$U^{gen}(\tilde{X}) := \{\tilde{\omega} \in \mathcal{KS}(\tilde{X}) | [\tilde{\omega}] \text{ is not contained in any multiple of a rational class}\}.$$

By Theorem 1.2, $\mathcal{KS}(\tilde{X})$ is open in $\mathcal{S}(\tilde{X})$. For any Kähler $\tilde{\omega} \in \mathcal{KS}(\tilde{X})$, we can perturb it to a form in $U^{gen}(\tilde{X})$, thus $U^{gen}(\tilde{X})$ is a dense subset of $\mathcal{KS}(\tilde{X})$.

Let $\tilde{\omega}, \tilde{\omega}' \in U^{gen}(\tilde{X}) \subset \mathcal{KS}(\tilde{X})$ and let $\{\tilde{\omega}_t | t \in [0,1]\}$ be a path in $\mathcal{KS}(\tilde{X})$ connecting $\tilde{\omega}$ and $\tilde{\omega}'$. Let $\eta_1, \cdots, \eta_{23}$ be closed 2-forms such that $\{[\eta_i] | i = 1, \cdots, 23\}$ is a basis of $H^2(\tilde{X}; \mathbb{R})$. Let $h(t) : [0,1] \to [0,1]$ be a non-constant smooth function such that $h(0) = h(1) = 0$, then there is a tuple of positive numbers $(v_1, ..., v_{23})$ such that

$$\{\tilde{\Omega}_t := \tilde{\omega}_t + h(t) \sum_{i=1}^{23} \epsilon_i \eta_i | t \in [0,1]\}, \quad |\epsilon_i| < v_i, i = 1, ..., 23$$

is a path of Kähler type forms connecting $\tilde{\omega}$ and $\tilde{\omega}'$. The paths are not contained in $U^{gen}(\tilde{X})$ only if

$$[\tilde{\Omega}_t] := [\tilde{\omega}_t] + h(t) \sum_{i=1}^{23} \epsilon_i [\eta_i] = sa$$

for some $s \in \mathbb{R}$ and $a \in H^2(\tilde{X}; \mathbb{Z})$, which determines a countable union of 2-dimensional sub-manifolds of $\{(\epsilon_1, \cdots, \epsilon_{23}) \in \mathbb{R}^{23}, |\epsilon_i| < v_i\}$. As a result, we can choose a path $\{\tilde{\Omega}_t | t \in [0,1]\}$ in $U^{gen}(\tilde{X})$ connecting $\tilde{\omega}$ and $\tilde{\omega}'$ in a 23-dimensional "slice" of $\mathcal{KS}(\tilde{X})$. This implies the (path) connectedness of $U^{gen}(\tilde{X})$. □

Proof of Theorem 1.6. This follows from Lemmas 3.31 and 3.32. □

4. The Q invariant for $\mathcal{A}_\omega$-noneffective ϵ, q_S and q_T

We first review the family Seiberg–Witten theory and introduce the relative family SW invariants in the canonical chamber. We then introduce the generalized Q invariant $Q_\epsilon : \pi_1(\mathcal{S}_\omega) \to \mathbb{Z}$ for $\mathcal{A}_\omega$-noneffective ϵ.

4.1. Relative family Seiberg–Witten invariants in the canonical chamber.

We first recall the definition and basic properties of family Seiberg–Witten invariant and we generally follow the notations in [29]. Then we introduce relative family Seiberg–Witten invariants in the canonical chamber.

4.1.1. *Family Seiberg–Witten invariant in an $\mathcal{L}$-chamber.* In this case, we consider a smooth fiber bundle $\mathcal{X} \to B$, where the fiber of the bundle is diffeomorphic to a connected, closed, smooth, oriented 4-dimensional manifold M and the base B is an oriented smooth manifold, possibly with nonempty boundary. Let $T(\mathcal{X}/B)$ be the vertical tangent bundle of $\mathcal{X}$. In order to define a family spin^c structure, we choose a family of metrics $\{g_b\}$ on $T(\mathcal{X}/B)$, which defines a principal SO(4) bundle of frames, Fr $\to M$.

A family spin^c structure $\mathcal{L}$ is a family of lifts of the frame bundle Fr to spin$^c(4)$ bundle. The group spin$^c(4)$ is isomorphic to $(\mathrm{SU}(2) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_2$, and there is a natural homomorphism from $\mathrm{SU}(2) \times \mathrm{U}(1)/\{\pm 1\}$ to U(2). So given a spin^c structure $\mathcal{L}$, we have two

natural associated U(2) bundles, S^+ and S^-. The Clifford multiplication from $\mathrm{T}^*(\mathcal{X}/B)\otimes S^+$ to S^- induces a family of quadratic maps $\sigma_b : \Gamma(S_b^+) \to i\Gamma(\Lambda_b^+)$. Let $L = det(S^+)$. A smooth family of U(1)-connections $A = \{A_b | b \in B\}$ on L and the family of Levi-Civita connections on $T(\mathcal{X}/B)$ determine a family of covariant derivatives on S^+ and the associated Dirac operators $D_{A_b} : \Gamma(S_b^{\pm}) \to \Gamma(S_b^{\mp})$.

Now we define the configuration space for $\mathcal{L}$ to be

$$\mathcal{C} = \{(A, \psi) | A_b \text{ is } U(1) \text{ connection on } L_b, \psi_b \in \Gamma(S_b^+), b \in B\},$$

Let $\mathcal{G}$ be a group bundle over B such that $\mathcal{G}_b$ is isomorphic to the gauge group $\mathrm{Map}(M, \mathrm{U}(1))$, which acts on $\mathcal{C}$ as follows: $u_b \cdot (\psi_b, A_b) = (e^{-if_b}\psi_b, A_b + 2idf_b)$, for $u_b = e^{-if_b} \in \mathcal{G}_b$. Clearly, $\mathcal{G}$ acts freely away from reducible pairs.

The parameter space for the family Seiberg–Witten equations on $\mathcal{X}/B$ is

$$\Gamma(\mathcal{X}) = \{(g, \mu) | \mu_b \text{ is } g_b - \text{self dual 2-form for every } b \in B\}. \tag{4.1}$$

When no confusion can arise, we abbreviate $\Gamma(\mathcal{X})$ to Γ. Each pair $(g, \mu) \in \Gamma$ defines a family Seiberg–Witten equation on $\mathcal{C}$:

$$\begin{cases} D_{A_b}\psi_b = 0, \\ \quad F_{A_b}^+ = \sigma_b(\psi_b) + i\mu_b. \end{cases} \tag{4.2}$$

Let $\mathfrak{sw}_{(g,\mu)}(\mathcal{L})$ be the space of solutions to (4.2). $\mathcal{G}$ preserves $\mathfrak{sw}_{(g,\mu)}(\mathcal{L})$ and the family moduli space is $\mathcal{M}_{(g,\mu)}^{\mathcal{L}} = \mathfrak{sw}_{(g,\mu)}(\mathcal{L})/\mathcal{G}$.

Definition 4.1. A solution $(A, \psi) \in \mathcal{C}$ is called irreducible if $\psi_b \neq 0$ for any b, and reducible otherwise.

- Let $\mathcal{WA}(\mathcal{L}) \subset \Gamma$ denote the wall crossing loci for $\mathcal{L}$, where $(g, \mu) \in \mathcal{WA}(\mathcal{L})$ if $\mathcal{M}_{(g,\mu)}^{\mathcal{L}}$ has a reducible solution.
- Let $\Gamma^*(\mathcal{L}) = \Gamma - \mathcal{WA}(\mathcal{L})$. A path connected component of $\Gamma^*(\mathcal{L})$ is called a $\mathcal{L}$-chamber.

We list some properties of the family moduli spaces.

Lemma 4.2. *Suppose $\mathcal{X} \to B$ is a smooth bundle with B compact and M as fibers with homology orientation specified. Let $\mathcal{L}$ be a family* spin^c *structure.*

(1) For any $(g, \mu) \in \Gamma$, $\mathcal{M}_{(g,\mu)}^{\mathcal{L}}$ is either empty or compact.

(2) Suppose $b^+(M) \geqslant 1$. For a generic pair $(g, \mu) \in \Gamma - \mathcal{WA}(\mathcal{L})$, $\mathcal{M}_{(g,\mu)}^{\mathcal{L}}$ is an oriented smooth manifold of dimension $d(\mathcal{L}) := \dim B + \frac{1}{4}(c_1(\mathcal{L})^2 - 2\chi(M) - 3\sigma(M))$, with its boundary lying over ∂B if $\partial B \neq \varnothing$.

(3) With a homology orientation of M, i.e. an orientation of $H^1(M; \mathbb{R}) \oplus H^2_+(M; \mathbb{R})$ specified, we get an orientation of $\mathcal{M}_{(g,\mu)}^{\mathcal{L}}$ when it is a smooth manifold.

Lemma 4.3. *Suppose that B is closed, $b^+(M) \geqslant 1$ and $\mathfrak{p} = (g, \mu)$ is generic in a $\mathcal{L}$-chamber.*

(1) If $\mathfrak{p}' = (g', \eta')$ is in the same $\mathcal{L}$-chamber and is also generic, then $\mathcal{M}_{\mathfrak{p}}^{\mathcal{L}}$ and $\mathcal{M}_{\mathfrak{p}'}^{\mathcal{L}}$ are cobordant as smooth closed, oriented manifolds.

(2) *Under the charge conjugation involution* $\mathcal{L} \to -\mathcal{L}$ *on the set of* spin^c *structures, there is a canonical diffeomorphism between* $\mathcal{M}^{\mathcal{L}}_{(g,\mu)}$ *and* $\mathcal{M}^{-\mathcal{L}}_{(g,-\mu)}$.

(3) *Suppose* $\mathcal{L}'$ *is a* spin^c *structure such that restrictions* $\mathcal{L}_b$ *and* $\mathcal{L}'_b$ *have equal Chern classes:* $c_1(\mathcal{L}_b) = c_1(\mathcal{L}'_b) \in H^2(M_b;\mathbb{Z})$. *If* $H^2(M;\mathbb{Z})$ *is torsion-free, then* $\mathcal{M}^{\mathcal{L}'}_{\mathfrak{p}}$ *is diffeomorphic to* $\mathcal{M}^{\mathcal{L}}_{\mathfrak{p}}$.

See [29][4][39] for details.

If the fiber of the group bundle $\mathcal{G}$ is replaced by $\mathcal{G}_0 := \{u \in \mathcal{G} | u_b(x_b) = id\}$, then we get a principal S^1-bundle $\mathfrak{sw}_{(g,\mu)}(\mathcal{L})/\mathcal{G}_0 \to \mathcal{M}^{\mathcal{L}}_{(g,\mu)}$ for a pair $(g,\mu) \in \Gamma^*(\mathcal{L})$. We define the hyperplane class to be the first Chern class of the S^1 bundle and denote it by u.

For any smooth family $\mathcal{X}/B$ and parameter family $\mathfrak{p} = (g,\mu)$ with $\mathcal{M}^{\mathcal{L}}_{(g_b,\mu_b)} = \varnothing$ for $b \in \partial B$, we define

$$\mathrm{fSW}(\mathcal{X}/B, \mathcal{L}; \mathfrak{p}) = \int_{\mathcal{M}^{\mathcal{L}}_{\mathfrak{p}}} u^{d/2}.$$

Definition 4.4. Suppose that B is closed and $b^+(M) \geqslant 1$. The family Seiberg–Witten invariant $\mathrm{fSW}(\mathcal{X}/B, \mathcal{L}, c) \in \mathbb{Z}$ in a $\mathcal{L}$-chamber c is defined as $\mathrm{fSW}(\mathcal{X}/B, \mathcal{L}; \mathfrak{p})$ where $\mathfrak{p}$ is generic in c.

4.1.2. *The canonical chamber* c_0 *when* B *is closed and simply connected.* Let $\mathcal{H}_g(M)$ be the space of g-harmonic 2-forms on M, we identify it with $H^2(M;\mathbb{R})$ by the Hodge theorem. Let $\mathcal{H}^+_g(X)$ denote the subspace of self-dual g-harmonic 2-forms and identify it with the corresponding subgroup of $H^2(M;\mathbb{R})$. Let H^+_g denote both the projection to $\mathcal{H}^+_g$ on the space of 2-forms and the projection on $H^2(M;\mathbb{R})$. Given a parameter family $(g,\mu) = \{(g_b,\eta_b) | b \in B\}$, since the cokernel of $d^+ : \Omega^1 \to \Omega^+$ is represented by $\mathcal{H}^+_{g_b}$, there is a reducible solution at b if and only if

$$H^+_{g_b}(\mu_b) = H^+_{g_b}(-iF_{A_b}).$$

Consider the period map

$$Per : \Gamma \to \mathcal{H}^+_b, \quad (g_b,\mu_b) \to H^+_{g_b}(\mu_b) + H^+_{g_b}(2\pi c_1(\mathcal{L})) \tag{4.3}$$

Since F_{A_b} is a closed 2-form representing the class $-2\pi i c_1(\mathfrak{s}_b)$, the existence of reducible solution is equivalent to

$$Per((g_b,\mu_b)) = 0. \tag{4.4}$$

From the description of $\mathcal{WA}(\mathcal{L}) \subset \Gamma$ by (4.4), $\mathcal{WA}(\mathcal{L})$ is a closed submanifold of codimension $\max(b^+(M) - \dim B, 0)$. In particular, $\Gamma^*(\mathcal{L})$ is path connected if $b^+(M) \geqslant \dim B + 2$. Furthermore, it is shown in [29] that $\Gamma - \mathcal{WA}(\mathcal{L})$ is homotopic to the space of sections of an S^{b^+-1} bundle over B. Let $\mathcal{H}(\mathcal{X})$ be the flat bundle of harmonic 2-forms over B, $\mathcal{H}^{pc}(\mathcal{X})$ the subbundle of positive square forms, and $\mathcal{H}^+(\mathcal{X})$ the subbundle of self-dual forms. By [Lemma 3.1, [33]], a choice of the homotopy S^{b^+-1} bundle is $\mathcal{H}^+(\mathcal{X}) - \{0_B\}$ or $\mathcal{H}^{pc}(\mathcal{X})$ and the period map

$$(g_b,\mu_b) \to Per((g_b,\mu_b)) \in \mathcal{H}^+_b - \{0\} \subset \mathcal{H}^{pc}_b. \tag{4.5}$$

When B is simply connected, the flat homology bundle $\mathcal{H}$ is trivial there is a canonical trivialization $\mathcal{H}(\mathcal{X}) \cong \mathcal{H}(M) \times B$. And we have the following result.

Lemma 4.5. *Suppose B is simply connected and closed, the chambers are in one-to-one correspondence with the homotopy classes of maps from B to S^{b^+-1}. More precisely, the chamber of $\mathfrak{p} = (g, \mu)$ is determined by the homotopy class of*

$$p_1 \circ Per : b \to Per(g_b, \mu_b) \in \mathcal{H}^+(X) \cong S^{b^+-1}.$$

Definition 4.6. When B is simply connected and $b^+(M) \geqslant 2$, the canonical chamber c_0 is defined to be the chamber when Per defines a trivial homotopy class.

Clearly, (a subset of) a contractible set of $\mathcal{H}^+(X)$ determines the canonical chamber. Given a positive square class a, a sufficient condition that (4.4) is violated is

$$\langle Per(g_b, \mu_b), a\rangle < 0, \quad b \in B. \tag{4.6}$$

Let $\Gamma^*(\mathcal{L}, a)$ be the "half-space" of pairs satisfying (4.6), then $\Gamma^*(\mathcal{L}, a)$ is contractible by [29] [33]. We define the corresponding chamber of $\mathfrak{p} \subset \Gamma^*(\mathcal{L}, a)$ to be c_a.

Lemma 4.7. *$c_a = c_0$ when $b^+(M) \geqslant 2$.*

Remark 4. In the presence of a symplectic structure ω, a natural choice is $a_b = [\omega]$.

4.1.3. *The relative family SW invariant* $r\mathrm{fSW}(M, \mathfrak{s}, \mathfrak{p})$. For a family B with boundary, if we assume that $\mathcal{M}^{\mathcal{L}}_{(g_b,\eta_b)} = \varnothing$ for $b \in \partial B$, we can also define the relative family Seiberg–Witten invariant.

We will consider the smoothly trivial family $B = \mathbb{D}^{2n}$ and $\mathcal{X} = M \times \mathbb{D}^{2n}$. In this case, restriction to a fiber of $\mathcal{X}$ induces a bijection between the family spin^c structures on $\mathcal{X}$ and the spin^c structures on M. And we can identify a parameter family $\mathfrak{p}$ of family Seiberg-Witten equations on $\mathcal{X}$ with a map from B to

$$\Gamma = \{(g, \mu) | g \text{ is a Riemannian metric on } M, \mu \text{ is a } g - \text{self dual 2-form}\}, \tag{4.7}$$

the parameter space of ordinary Seiberg-Witten equation. Let $\mathcal{L}$ be the family spin^c structure on $\mathcal{X}$ corresponding to $\mathfrak{s}$, for simplicity of writing, we define

$$\mathrm{fSW}(M, \mathfrak{s}, \mathfrak{p}) = \mathrm{fSW}(\mathcal{X}/\mathbb{D}^{2n}, \mathcal{L}, \mathfrak{p}).$$

Consider a spin^c structure $\mathfrak{s}$ on M with

$$d(\mathfrak{s}) = \frac{1}{4}\big(c_1(\mathfrak{s})^2 - 2\chi(M) - 3\sigma(M)\big) = -2n.$$

Kronheimer studies the case $b^+(M) > 2n+1$ in [§ 2, [26]], where there is a unique chamber.

Let $\mathcal{WA}(\mathfrak{s}), \Gamma^*(\mathfrak{s})$ and $\Delta(\mathfrak{s})$ be defined similar as Definition 4.1.

Definition 4.8. Let $\gamma : (S^{2n-1}, *) \to (\Gamma^*(\mathfrak{s}), p)$ be a based map such that

$$\gamma(t) = (g_t, \eta_t) \in \Gamma^*(\mathfrak{s}, a) \backslash \Delta_{\mathfrak{s}}, \quad t \in S^{2n-1}$$

for some positive class $a \in H^2(M; \mathbb{R})$. We use $\mathcal{L}$ to denote the family spin^c structure on $\mathcal{X}$ corresponding to $\mathfrak{s}$, and define the relative invariant in the canonical chamber

$$r\mathrm{fSW}(M, \mathfrak{s}, \gamma, a) := \mathrm{fSW}(M, \mathfrak{s}, \tilde{\gamma}),$$

where $\tilde{\gamma}$ is an extension of γ such that $\tilde{\gamma}(b) = (g_b, \eta_b) \in \Gamma^*(\mathfrak{s}, a)$ for $b \in \mathbb{D}^{2n}$.

Lemma 4.9. *$r\mathrm{fSW}(M,\mathfrak{s},\gamma,a)$ only depends on the homotopy class of*

$$[\gamma]\in\pi_{2n-1}(\Gamma^*(\mathfrak{s},a)\backslash\Delta_{\mathfrak{s}}),$$

and it gives rise to a homomorphism $\pi_{2n-1}(\Gamma^*(\mathfrak{s},a)\backslash\Delta_{\mathfrak{s}})\to\mathbb{Z}$.

Proof. Different choices of extension $\tilde{\gamma}$ differ by a S^{2n} family $\mathfrak{p}$ in $\gamma^*(\mathfrak{s},a)$. We also use $\mathcal{L}$ to denote the family spin^c structure, and the relative family Seiberg–Witten invariant over $\int_{\mathcal{M}^{\mathcal{L}}_{\mathfrak{p}}} u^{-d(\mathcal{L})/2}$ is trivial. Let γ' be another based map $S^{2n-1}\to\Gamma^*(\mathfrak{s})$ and $\tilde{\gamma}'$ is an extension of γ' in $\Gamma^*(\mathfrak{s})$, then

$$r\mathrm{fSW}(M,\mathfrak{s},\gamma\vee\gamma',a)=\mathrm{fSW}(M,\mathfrak{s},\tilde{\gamma}\vee\tilde{\gamma}')=\mathrm{fSW}(M,\mathfrak{s},\tilde{\gamma})+\mathrm{fSW}(M,\mathfrak{s},\tilde{\gamma}'),$$

which is equal to $r\mathrm{fSW}(M,\mathfrak{s},\gamma,a)+r\mathrm{fSW}(M,\mathfrak{s},\gamma',a)$. This proves the additivity. □

4.2. **The generalized invariant $Q_\epsilon:\pi_1(\mathcal{S}_\omega)\to\mathbb{Z}$ for $\mathcal{A}_\omega$-noneffective ϵ.** Now we begin to discuss some properties of Seiberg–Witten theory for symplectic 4-manifolds. Let (M,ω) be a closed connected oriented symplectic 4-manifold. There is a natural homology orientation, called the symplectic homology orientation.

We pick an almost Kähler 3-tuple (ω,g,J), where J is compatible with ω and g is determined by ω and J. In this case, for a class $\epsilon\in H^2(M;\mathbb{Z})$, we have a spin^c structure $\mathfrak{s}_\epsilon$ on M with $c_1(\mathfrak{s}_\epsilon)=c_1(M,\omega)+2\epsilon$, and the formal dimension of the Seiberg–Witten moduli space as a real manifold is:

$$d(\mathfrak{s}_\epsilon)=\langle\epsilon,\epsilon\rangle-\langle K,\epsilon\rangle. \tag{4.8}$$

Following [42], a "large perturbation" $\eta=-iF^+_{A_0}-\rho\omega$ is a real self-dual 2-form for each almost Kähler surface.

4.2.1. *$\mathcal{A}_\omega$-noneffective classes.* Recall that $\mathcal{S}_\omega$ is the space of symplectic forms isotopic to ω and $\mathcal{A}_\omega$ is the space of almost complex structures compatible with some symplectic form in $\mathcal{S}_\omega$.

Definition 4.10. We say that a nonzero class $\epsilon\in H^2(M;\mathbb{Z})$ is $\mathcal{A}_\omega$-noneffective if for any $J\in\mathcal{A}_\omega$, $\mathrm{PD}(\epsilon)$ is not represented by a non-constant J-holomorphic curve.

For any J tamed by ω, if there is a J-holomorphic subvariety in $\mathrm{PD}(\epsilon)$, then $\langle\omega,\epsilon\rangle>0$, thus the following Lemma holds.

Lemma 4.11. *When $\langle\omega,\epsilon\rangle\leqslant 0$, ϵ is $\mathcal{A}_\omega$-non-effective.*

Lemma 4.12. *Let $(\tilde{M},\tilde{\omega})$ be the one point blowup of a minimal symplectic 4-manifold (M,ω) and e the exceptional class. For any $n\in\mathbb{Z}$ and any class $\xi\in H_2(M;\mathbb{Z})$ with $\omega(\xi)\geqslant 0$, $ne-\xi$ is $\mathcal{A}_\omega-$noneffective.*

Proof. When $n\leqslant 0$, the statement follows from Lemma 4.11. Now suppose that $n>0$. By [Lemma 2.3, [9]], e is represented by an embedded pseudo-holomorphic curve C_0 in $(\tilde{M},\tilde{J}_t)$. Since $(ne-\gamma)\cdot e=-1<0$, any $\tilde{J}$-holomorphic curve in the class $ne-\xi$ must have C_0 as a component, and a curve in the class $(n-1)e-\xi$ exists. Continuing the procedure, we get the existence of a curve in the class $-\xi$. This is impossible by Lemma 4.11. As a result, for any $\tilde{J}\in\mathcal{A}_{\tilde{\omega}}$, there will be no $\tilde{J}$-holomorphic curve representing $ne-\xi$. □

Let $\alpha \in \pi_1(\mathcal{S}_\omega, \omega)$ and $\{\omega_t | t \in S^1\}$ be a representative of α. Then we choose a smooth S^1 family of almost complex structures $\{J_t | t \in S^1\}$ fiberwise compatible with $\{\omega_t\}$ and $\{g_t = \omega_t(\cdot, J_t \cdot) | t \in S^1\}$ is a family of almost Kähler metrics. Choose a disk family $\{g_b | b \in \mathbb{D}^2\}$ of Riemannian metrics extending $\{g_t | t \in S^1\}$.

We regard $\{g_b | b \in \mathbb{D}^2\}$ as a family of fiberwise metrics on the trivial bundle $\mathcal{X} = M \times D$, and let $\mathcal{L}_\epsilon$ be a family spin^c structure of $\mathrm{T}(\mathcal{X}/\mathbb{D}^2)$ such that ϵ is $\mathcal{A}_\omega$-noneffective. Following the method of Taubes, we define

$$\eta(t, \rho) := -iF^+_{A_{0_t}} - \rho\omega_t. \tag{4.9}$$

We choose $\rho > 0$ large enough so that $\langle Pr(g_t, \eta(t, \rho)), \omega_t \rangle < 0$.

By Taubes' SW⇒Gr [Theorem 1.3, [42]], we have

Theorem 4.13. *Suppose $\epsilon \neq 0$ and ϵ is $\mathcal{A}_\omega$-noneffective, then there are no solutions to* (4.2) *parametrized by*

$$\mathfrak{T}_\rho(\omega, J) := (\omega(\cdot, J\cdot), \eta(t, \rho))$$

for ρ large.

Lemma 4.14. *For an $\mathcal{A}_\omega$-noneffective class ϵ, when ρ is big enough, the corresponding moduli space $\mathcal{M}^{\mathfrak{s}_\epsilon}_{g_t, \eta(t,\rho)}$ is empty for each $t \in \partial\mathbb{D}^2$. Hence the map*

$$\mathfrak{T}_\rho : (\omega, J) \to (\omega(\cdot, J\cdot), \eta(t, \rho))$$

induces a homomorphism $\mathcal{T} : \pi_1(\mathcal{S}_\omega, \omega) \to \pi_1(\Gamma^*(\mathfrak{s}_\epsilon) \backslash \Delta_{\mathfrak{s}_\epsilon}; \mathbb{Z})$.

Proof. By a compact family version of Theorem 4.13, if there is a sequence $\rho^i \to \infty$ such that solution of $\mathfrak{sw}_{g_{t_i}, \eta_{t_i}, \rho^i}(\mathcal{L}_\epsilon) \neq \varnothing$, then there will be a J_t-holomorphic PD(ϵ)-curve if a subsequence of $\{t_i\}$ converges to t, which is impossible. This proves the first statement.

Consequently, $\mathfrak{T}_\rho(\omega_t) \in \Gamma^*(\mathfrak{s}_\epsilon) \backslash \Delta_{\mathfrak{s}_\epsilon}$ for each $t \in S^1$ and sufficiently large ρ. Since the space of almost complex structures compatible with ω is contractible, for different loops $\{J_t\}, \{J'_t\}$ fiberwise compatible with $\{\omega_t\}$, we can take a cylinder family

$$\{J_{r,t} | J_{r,t} \text{ is compatible with } \omega_t\}$$

connecting them. When ρ is big enough, the loops $\{\mathfrak{T}_\rho(\omega_t, J_t)\}$ and $\{\mathfrak{T}_\rho(\omega_t, J'_t)\}$ will be homotopic in $\Gamma^*(\mathfrak{s}_\epsilon) \backslash \Delta_{\mathfrak{s}_\epsilon}$. For a similar reason, the homotopy class we get does not depend on the choices of representatives of symplectic loops. Thus $\mathcal{T}$ is a well-defined map. Then it will be easy to see $\mathcal{T}$ is a well-defined homomorphism. □

Remark 5. By the proof of Lemma 4.5, we can see that the image of $\mathfrak{T}_\rho$ will be contained in a half-space $\Gamma^*(\mathcal{L}, [\omega])$, thus must bound a disk, even in the case when $b^+(M) = 2$.

4.2.2. *The Q invariant and properties.* Let (M, ω) be a closed symplectic 4-manifold with $b^+ \geqslant 1$. For $\alpha \in \pi_1(\mathcal{S}_\omega, \omega)$, let $\ell = \{\omega_t | t \in S^1\}$ be a representative of α and let $\{J_t\}$ to be a loop of almost complex structures s.t. J_t is compatible to ω_t. Suppose ϵ is $\mathcal{A}_\omega$-noneffective and $d(\epsilon) = \epsilon^2 - \langle K, \epsilon \rangle = -2$. The map

$$\gamma : (S^1, *) \to (\Gamma^*(\mathfrak{s}_\epsilon), p), \quad \gamma(t) = \mathfrak{T}_\rho(\omega_t, J_t)$$

is contained in $\in \Gamma^*(\mathfrak{s}_\epsilon, [\omega]) \backslash \Delta_{\mathfrak{s}_\epsilon}$ when ρ is large enough.

Definition 4.15. Suppose that ϵ is $\mathcal{A}_\omega$-noneffective and let $d(\epsilon) = \epsilon^2 - \langle K, \epsilon\rangle = -2$. The Kronheimer invariant $Q_\epsilon : \pi_1(\mathcal{S}_\omega, \omega) \to \mathbb{Z}(\mathbb{Z}_2)$ is defined as

$$Q_\epsilon(\alpha) = Q(\ell) = r\mathrm{fSW}(M, \mathfrak{s}_\epsilon, \gamma, [\omega]).$$

Remark 6. The $[\omega]$-negative version of Q_ϵ was originally introduced in [26] under the assumption $b^+ > 3$. Smirnov observed that it can still be defined and have the same homomorphism property when $b^+ = 3$. This observation is crucial since $b^+ = 3$ for K3 surfaces. It is further observed in [33] that Smirnov's extension is still valid as long as $b^+ \geqslant 1$. Our $\mathcal{A}_\omega$-noneffective version is defined when $b^+ \geqslant 1$.

As in section 4.1.3, the extension of γ is a family $\tilde{\gamma} = \{(g_b, \eta_b)|b \in D^2\}$ such that

$$\langle Per((g_b, \eta_b)), \omega\rangle < 0, \quad \forall b \in \mathbb{D}^2, \tag{4.10}$$

if $\tilde{\gamma}$ satisfies the inequality above, we call it an admissible extension [39]. In this paper, we actually compute the Kronheimer invariant using a disk family of symplectic forms.

Definition 4.16. Let $D = \{\omega_b|b \in \mathbb{D}^2\}$ be a smooth family of symplectic forms such that $\partial D \subset \mathcal{S}_\omega$. We say D is ω-admissible if

$$\langle [\omega_b], [\omega]\rangle > 0, \quad \forall b \in \mathbb{D}^2. \tag{4.11}$$

Lemma 4.17. *Let $\ell = \{\omega_t|t \in S^1\} \subset \mathcal{S}_\omega$ be a smooth loop family of symplectic forms, and ℓ bounds an $\omega-$admissible symplectic disk family $D = \{\omega_b|b \in \mathbb{D}^2\}$ in $\mathcal{S}(M)$.*

- *Let $\{J_b|b \in \mathbb{D}^2\}$ be an almost complex family such that J_b is compatible with ω_b, and let $\tilde{\gamma}_\rho = \{\mathfrak{T}_\rho(\omega_b, J_b)|b \in \mathbb{D}^2\}$. Then $\mathrm{fSW}(M, \mathfrak{s}_\epsilon, \tilde{\gamma}_\rho)$ is equal to $Q_\epsilon(\ell)$ when ρ is big enough.*
- *Suppose there is a neighborhood U of $\omega \in \mathcal{S}(M)$ such that ϵ is $\mathcal{A}_{\omega'}$-noneffective for any $\omega' \in U$. If a loop $l' \subset \mathcal{S}_{\omega'}$ is close enough to ℓ, then we have $Q_\epsilon(l) = Q_\epsilon(l')$.*
- *If ϵ is $\mathcal{A}_{\omega_b}$-noneffective for each $b \in \mathbb{D}^2$, then $Q_\epsilon(l) = 0$.*
- *Q_ϵ is independent of the choices of admissible extensions, and Q_ϵ defines a homomorphism from $\pi_1(\mathcal{S}_\omega, \omega)$ to $\mathbb{Z}$ (or $\mathbb{Z}_2$).*

Proof. For the first result, it's easy to see $\langle [\omega_b], [\omega]\rangle > 0$ implies that (4.10) holds when ρ is big enough.

For the second statement, set $\ell' = \{\omega'_t|t \in S^1\}$. Since ℓ' is close enough to ℓ, there is a cylinder family $A(\ell, \ell') := \{\omega_{s,t}|s \in [0,1], t \in S^1\}$ of symplectic forms from ℓ to ℓ' such that

$$\langle \omega_b, \omega'\rangle > 0, \quad \forall b \in \mathbb{D}^2, \qquad \langle \omega', \omega_{s,t}\rangle > 0, \quad \forall (s,t) \in [0,1] \times S^1.$$

Since $\omega_t = \omega_{0,t}$ for each $t \in S^1$, $D' := D \cup A(\ell, \ell')$ is ω'-admissible disk family.

Let $\{J_{s,t}|(s,t) \in [0,1] \times S^1\}$ be a family of almost complex structures s.t. $J_{s,t}$ is compatible with $\omega_{s,t}$ and $J_{0,t} = J_t$ for each $t \in S^1 \cong \partial\mathbb{D}^2$, and set $\mathfrak{a}_\rho := \{\mathfrak{T}_\rho(\omega_{s,t}, J_{s,t})|(s,t) \in [0,1] \times S^1\}$. When ρ is sufficiently large, we can see that

$$\mathrm{fSW}(M, \mathfrak{s}_\epsilon, \tilde{\gamma}_\rho) = \mathrm{fSW}(M, \mathfrak{s}_\epsilon, \tilde{\gamma}_\rho \cup \mathfrak{a}_\rho) \tag{4.12}$$

since there does not exist $J_{s,t}$-holomorphic PD(ϵ)-curve for each $(s,t) \in [0,1] \times S^1$. Since $Q_\epsilon(\ell)$ is equal to left hand side of (4.12) and $Q_\epsilon(\ell')$ is equal to the right hand side of (4.12), we prove the conclusion.

The third result is a direct corollary of [Theorem 1.3, [42]]. And the fourth result is a direct corollary of Lemma 4.9 and Lemma 4.14. □

For a Kähler disc family equipped with the parameters introduced by Taubes, we can compute the (family) Seiberg Witten invariant using the following Theorem.

Theorem 4.18 (Chapter 7 [32])**.** *For sufficiently large ρ, there are no solutions to* (4.2) *parametrized by $\mathfrak{T}_\rho$ if $\epsilon \notin H^{1,1}(M;\mathbb{R})$.*

When $\epsilon \neq 0 \in H^{1,1}(M;\mathbb{R})$ and $\langle [\omega], \epsilon\rangle > 0$, for ρ big enough, the solutions to (4.2) *parametrized by $\mathfrak{T}_\rho$ are irreducible and, modulo gauge transformations, are in one-to-one correspondence with the set of fiberwise holomorphic curves (possibly non-reduced) whose fundamental class is Poincaré dual to ϵ.*

Lemma 4.19. *Let ϵ, ℓ and D be as in definition 2.1 and the ϵ-Hodge type twist is spherical or elliptic. Then $Q_\epsilon(l) = \pm 1$.*

Proof. Over a Kähler family $D = \{(\omega_b, J_b)|b \in \mathbb{D}^2\}$, set $\mathfrak{p}_\rho = \{\mathfrak{T}_\rho(\omega_b, J_b)|b \in \mathbb{D}^2\}$. Since $\{\omega_b|b \in \mathbb{D}^2\}$ is ω_1-admissible, $Q_\epsilon(\ell)$ is equal to $\mathrm{fSW}(M, \mathfrak{s}_\epsilon, \mathfrak{p}_\rho)$ for sufficiently large ρ. By the intersection positivity, the curve in $\mathrm{PD}(\epsilon)$ is unique in the central fiber, Since $\epsilon \in H^{1,1}(M; J_b; \mathbb{R})$ if and only if $b = 0$, the curve is unique in $\mathrm{PD}(\epsilon)$ over the disk family. And by Theorem 4.18, there is a unique solution of the family Seiberg–Witten equation on central fiber. In the specific cases, the identification between curve counting in $\mathrm{PD}(\epsilon)$ and isolated solutions of $\bigcup_{b\in\mathbb{D}^2} \mathcal{M}^{\mathfrak{s}_\epsilon}_{\mathfrak{T}_\rho(\omega_b,J_b)}$ over Kähler families has been proved in [18][26][40]. □

4.3. **The spherical and elliptic q homomorphisms on $K(\tilde{X}, \tilde{\omega})$.** In this section, we first let (M, ω) be a closed symplectic manifold such that $H^2(M;\mathbb{Z})$ is torsion free and $b_1(M) = 0$.

We introduce the associated q invariant $q_\epsilon : K(M,\omega) \to \mathbb{Z}_2$ when both Q_ϵ and $Q_{K-\epsilon}$ are defined.

Lemma 4.20. *Suppose that Q_ϵ and $Q_{K-\epsilon}$ are both defined on $\pi_1(\mathcal{S}_\omega, \omega)$. Let $\ell = \{\omega_t|t \in S^1\}$ be a based loop in $\mathcal{S}_\omega$ such that $\omega_t = f_t^*\omega_0$ for a loop of diffeomorphisms $f = \{f_t \in \mathrm{Diff}_0(M)|f_1 = id\}$. Then $Q_\epsilon(\ell) = Q_{K-\epsilon}(\ell) \pmod 2$.*

Proof. Following Smirnov[40], we set

$$J_t = (f_t^{-1})_* \circ J \circ (f_t)_*, \quad g_t = g \circ (f_t)_*, \quad \eta_t = f^*\eta_0.$$

Let $\mathcal{L}_\epsilon, \mathcal{L}_{K-\epsilon}$ be the family spin^c structure determined by $\mathfrak{s}_\epsilon, \mathfrak{s}_{K-\epsilon}$, we take admissible extensions $\tilde{\gamma}_\rho = \{(g_b, \eta_b)|b \in \mathbb{D}^2\}$ of $\gamma_\rho = \{\mathfrak{T}_\rho(\omega_t, J_t)|t \in S^1\}$. Let

$$E_f := \mathbb{D}^2_+ \times M \bigcup \mathbb{D}^2_- \times M/\sim, \quad (t_+, x) \sim (t_-, f_t(x)).$$

Since $b_1(M) = 0$, $\mathcal{L}_\epsilon$ and $\mathcal{L}_{K-\epsilon}$ extend to family spin^c structures on E_f, and since $H^2(M;\mathbb{Z})$ is torsion free, the family SW-invariants are independent of the choices of extensions [39]. Thus, we can still denote them by $\mathcal{L}_\epsilon, \mathcal{L}_{K-\epsilon}$, respectively. The admissible family $\tilde{\gamma}_\rho$ on $\mathbb{D}^2_+$, together with trivial parameter family $\{(g_0, \eta_0)|b \in \mathbb{D}^2_-\}$, define a parameter family

$\mathfrak{p} = (g, \eta) = \{(g_z, \eta_z) | z \in S^2\}$ in both $\Gamma^*(\mathcal{L}_\epsilon)$ and $\Gamma^*(\mathcal{L}_{K-\epsilon})$. The corresponding family Seiberg–Witten invariants

$$\mathrm{fSW}(E_f/S^2, \mathcal{L}_\epsilon, \mathfrak{p}), \quad \mathrm{fSW}(E_f/S^2, \mathcal{L}_{K-\epsilon}, \mathfrak{p})$$

are equal to $Q_\epsilon(\ell)$ and $Q_{K-\epsilon}(\ell)$, respectively.

By the definition of admissible parameter family, $Pr((g_z, \eta_z)) \in \mathbf{K}_{-[\omega]}$ for each $z \in S^2$, thus $Pr((g_z, -\eta_z)) \in \mathbf{K}_{[\omega]}$. By the "charge" conjugation symmetry of family Seiberg–Witten invariants,

$$\mathrm{fSW}(E_f, \mathcal{L}_\epsilon, (g, \eta)) = \mathrm{fSW}(E_f, \mathcal{L}_{K-\epsilon}, (g, -\eta)) = \mathrm{fSW}(E_f, \mathcal{L}_{K-\epsilon}, (g, \eta)) \pmod 2,$$

which completes the proof. □

Suppose that Q_ϵ and $Q_{K-\epsilon}$ are both defined on $\pi_1(\mathcal{S}_\omega, \omega)$. We have a homomorphism $q_\epsilon = Q_\epsilon - Q_{K-\epsilon} : K(M, \omega) \to \mathbb{Z}_2$.

Now let X be a K3 surface and $\tilde{X}$ be its blowup and $\tilde{\omega}$ is a symplectic form on $\tilde{X}$. By Lemma 4.12, for each $\gamma \in S(X, \kappa(\tilde{\omega}))$, both γ and $e - \gamma$ are $\mathcal{A}_{\tilde{\omega}}$-non-effective, for each $\delta \in T_\lambda(X, \kappa(\tilde{\omega}))$, both $\delta - e$ and $2e - \delta$ are $\mathcal{A}_{\tilde{\omega}}$-non-effective.

Definition 4.21. We have spherical and elliptic q homomorphisms for γ and δ respectively,

$$q_\gamma := Q_\gamma - Q_{e-\gamma}, \quad q_{\delta-e} := Q_{\delta-e} - Q_{2e-\delta} : \quad K(\tilde{X}, \tilde{\omega}) \to \mathbb{Z}_2.$$

Remark 7. By [33], we can also define an integer valued q-invariant.

Theorem 1.7 is proved by the following 4 lemmas, corresponding to the 4 items.

Lemma 4.22. *For $\delta \in T_{\lambda(\tilde{\omega})}(X, \kappa(\tilde{\omega}))$, $q_{\delta-e}$ is defined on $K(\tilde{X}, \tilde{\omega})$. For any loop $l_{\tilde{\omega}} \subset \mathcal{S}_{\tilde{\omega}}$ such that $q_{\delta-e}(\partial_* \ell_{\tilde{\omega}})$ is non-trivial, then there is a neighborhood $U \subset \mathcal{KS}(\tilde{X})$ of $\tilde{\omega}$ such that $q_{\delta-e}$ is defined for each $\tilde{\omega}' \in U$, and there is a loop $\ell_{\tilde{\omega}'} \subset \mathcal{S}_{\tilde{\omega}'}$ which defines a non-trivial symplectic mapping class.*

Proof. Let $D = \{\tilde{\omega}_b | b \in \mathbb{D}^2\} \subset \mathcal{S}_{\tilde{\omega}}$ be the $\tilde{\omega}$-admissible disk family extending $\ell_{\tilde{\omega}}$ and let ℓ_J be a loop of almost complex structures fiberwise compatible with $\ell_{\tilde{\omega}}$. Since ℓ_J is compact, we can take a small neighborhood U of $\tilde{\omega}$ in $\mathcal{S}(\tilde{X})$ such that $\forall \tilde{\omega}' \in U$,

(1) $\langle \tilde{\omega}', \epsilon \rangle = \langle \tilde{\omega}', \delta - e \rangle < 0$,
(2) $\langle \tilde{\omega}', \delta \rangle > 0$,
(3) $\langle \tilde{\omega}', \tilde{\omega}_b \rangle > 0$,
(4) there is a loop $\ell_{\tilde{\omega}'}$ in $\mathcal{S}_{\tilde{\omega}'}$ which fiberwise tames ℓ_J.

By the first two items, $q_{\delta-e}$ is well-defined on $K(\tilde{X}, \tilde{\omega}')$. Since the space of symplectic forms tamed by an almost complex structure is contractible, there is a linear homotopy

$$H : [0, 1] \times S^1 \to \mathcal{S}, \quad H(s, t) = (1 - s)\ell_{\tilde{\omega}} + s\ell_{\tilde{\omega}'}(t).$$

By our assumption on $\tilde{\omega}$ and $\tilde{\omega}'$, we have

(1) $\langle H(s, t), \delta - e \rangle < 0$,
(2) $\langle H(s, t), \delta \rangle > 0$,
(3) $\langle H(s, t), \tilde{\omega}' \rangle > 0$.

By the first two assumptions for $H(s,t)$, both $\delta - e$ and $2e - \delta$ are $\mathcal{A}_{H(s,t)}$-non-effective. Let $\{J_b | b \in \mathbb{D}^2\}$ be an almost complex family s.t. J_b is compatible to $\tilde{\omega}_b$, and let $\{J_{s,t} | s \in [0,1], t \in S^1\}$ be an almost complex family s.t. $J_{s,t}$ is compatible to $H(s,t)$ such that $J_{0,t} = J_t$. Set $\mathfrak{p}_\rho = \{\mathfrak{T}_\rho(\tilde{\omega}_b, J_b)\}$ and $\mathfrak{a}_\rho = \{\mathfrak{T}_\rho(H(s,t), J_{s,t})\}$, then

$$FSW(\tilde{X}, \mathfrak{s}_\epsilon, \mathfrak{p}_\rho \cup \mathfrak{a}_\rho) = \mathrm{fSW}(\tilde{X}, \mathfrak{s}_\epsilon, \mathfrak{p}_\rho) = Q_\epsilon(\ell_{\tilde{\omega}}),$$

$$\mathrm{fSW}(\tilde{X}, \mathfrak{s}_{K-\epsilon}, \mathfrak{p}_\rho \cup \mathfrak{a}_\rho) = \mathrm{fSW}(\tilde{X}, \mathfrak{s}_{K-\epsilon}, \mathfrak{p}_\rho) = Q_{K-\epsilon}(\ell_{\tilde{\omega}})$$

when ρ is large enough. Since

$$\langle \tilde{\omega}', \tilde{\omega}_b \rangle > 0, \quad \langle H(s,t), \tilde{\omega}' \rangle > 0,$$

$D \cup \{H(s,t)\}$ is a $\tilde{\omega}'$-admissible disk family of symplectic forms, then

$$Q_\epsilon(\ell_{\tilde{\omega}'}) = FSW(\tilde{X}, \mathfrak{s}_\epsilon, \mathfrak{p}_\rho \cup \mathfrak{a}_\rho) = Q_\epsilon(\ell_{\tilde{\omega}}), \quad Q_{K-\epsilon}(\ell_{\tilde{\omega}'}) = \mathrm{fSW}(\tilde{X}, \mathfrak{s}_{K-\epsilon}, \mathfrak{p}_\rho \cup \mathfrak{a}_\rho) = Q_{K-\epsilon}(\ell_{\tilde{\omega}}).$$

Suppose that $q_{\delta-e}(\partial_* \ell_{\tilde{\omega}}) = 1 \in \mathbb{Z}_2$ for some $\ell_{\tilde{\omega}} \subset \mathcal{S}_{\tilde{\omega}}$. Then $q_{\delta-e}(\partial_* \ell_{\tilde{\omega}'}) = q_{\delta-e}(\partial_* \ell_{\tilde{\omega}}) = 1$. So $q_{\delta-e}$ is also nontrivial on $K(M, \tilde{\omega}')$. □

Lemma 4.23. *$q_{\delta-e}$ vanishes on squared Seidel-Dehn twists.*

Proof. By Lemma 2.4, there is a disk family $D = \{\omega_b | b \in \mathbb{D}^2\}$ in $S(\tilde{X})$ with the change of cohomology classes of elements in D small enough, such that

(1) The monodromy of ∂D is isotopic to τ_L^2,
(2) $\langle \omega_b, \delta - e \rangle < 0$, $\forall b \in \mathbb{D}^2$,
(3) $\langle \omega_b, \omega \rangle > 0$, $\forall b \in \mathbb{D}^2$.

The second and third conditions since $\langle \omega, \delta - e \rangle < 0$ and $\omega^2 > 0$.

By Theorem 4.17, $Q_\epsilon(D)$ is equal to $\mathrm{fSW}(\tilde{X}, \mathfrak{s}_{\delta-e}, \{\mathfrak{T}_\rho(\omega_b, J_b) | b \in \mathbb{D}^2\})$, where $\{J_b\}$ is any family of almost complex structures such that J_b is compatible with ω_b, $\forall b \in \mathbb{D}^2$. Since $\langle \omega_b, \delta - e \rangle < 0$ for each $b \in \mathbb{D}^2$, there does not exist J_b-holomorphic PD$(\delta - e)$-curve for $b \in \mathbb{D}^2$. Thus $Q_{\delta-e}(\partial D) = 0$ by Lemma 4.17. By the proof of Lemma 4.12, since $\langle \omega_b, \delta \rangle > 0$ on each fiber of D, there does not exist J_b-holomorphic PD$(2e - \delta)$-curve, and we can get $Q_{2e-\delta}(\partial D) = 0$. So $q_{\delta-e}$ vanishes on each generator of $DS(\tilde{X}, \tilde{\omega})$, and $q_{\delta-e} \equiv 0$ on $DS(\tilde{X}, \tilde{\omega})$. □

Now we assume that $\tilde{\omega}$ is a Kähler type form.

Lemma 4.24. *$q_S = \oplus q_\gamma$ is nontrivial on spherical Hodge type twists, and $q_T = \oplus q_{\delta-e}$ is nontrivial on elliptic Hodge type twists.*

Proof. Recall the construction of γ-Hodge type twist for $\gamma \in \bar{S}(X, \kappa)$ in section 3.4.2. Let $\{(\tilde{\omega}_b, J_b) | b \in \mathbb{D}^2\}$ be the disk family of Kähler structures such that $\tilde{\omega}_1 = \tilde{\omega}$ and set $\mathfrak{p}_\rho = \{\mathfrak{T}_\rho(J_b, \omega_b) | b \in \mathbb{D}^2\}$. Let $D_\epsilon = \{\tilde{\omega}_b | b \in \mathbb{D}^2\}$, let $\ell_\gamma = \{\tilde{\omega}_t | t \in S^1\}$ its boundary and define $\phi_\gamma = \partial_* \ell_\gamma$ to be its monodromy. Since D_ϵ is $\tilde{\omega}$-admissible, $Q_{e-\gamma}(\ell_\gamma)$ is equal to $\mathrm{fSW}(\tilde{X}, \mathfrak{s}_{e-\gamma}, \mathfrak{p}_\rho)$ for ρ big enough. Since $\gamma \notin H^{1,1}(\tilde{X}, J_b; \mathbb{R})$ for each $b \in \mathbb{D}^2 \backslash \{0\}$, there is no curve in PD$(e - \gamma)$ on non-central fibers of D_ϵ. If there is a J_0-holomorphic PD$(e - \gamma)$-curve, then $-$PD(γ) is represented by a curve on its minimal model, which is impossible because of the existence of a PD(γ)-curve. Then by [Theorem 1.3, [42]],

$Q_{e-\gamma}(\ell_\gamma) = \mathrm{fSW}(\tilde{X}, \mathfrak{s}_{e-\gamma}, \mathfrak{p}_\rho) = 0$ when ρ is large enough. Then $q_\gamma(\phi_\gamma) = 1$ by Lemma 4.19, and the spherical Hodge type twist is nontrivial.

For a $(\delta - e)$-Hodge type twist constructed in section 3.4.2, let $\{(\tilde{\omega}_b, J_b)|b \in \mathbb{D}^2\}$, $\ell_{\delta-e}$, $\phi_{\delta-e}$, $D_{\delta-e} = \{(\tilde{\omega}_b, J_b)|b \in \mathbb{D}^2\}$ and $\mathfrak{p}_\rho$ be defined similarly. Similar to the spherical case, $Q_{2e-\delta}(\ell) = \mathrm{fSW}(\tilde{X}, \mathfrak{s}_{2e-\delta}, \mathfrak{p}_\rho)$ for ρ big enough. By the proof of Lemma 4.12, since $\langle \tilde{\omega}_b, \delta \rangle \geqslant 0$ for each b, there does not exist J_b-holomorphic $\mathrm{PD}(2e-\delta)$-curve for each $b \in \mathbb{D}^2$. As a result, $Q_{2e-\delta}(\ell) = \mathrm{fSW}(\tilde{X}, \mathfrak{s}_{2e-\delta}, \mathfrak{p}_\rho) = 0$. Again, by Lemma 4.19, $q_{\delta-e}(\partial\ell) = 1$ and the elliptic Hodge type twist is nontrivial. □

Lemma 4.25. *When $\lambda < Se(\kappa)$, the homomorphism*

$$q_S \oplus q_T := \bigoplus_{\bar{\gamma} \in \bar{S}(X,\kappa)} q_{\bar{\gamma}} \oplus \bigoplus_{\delta \in T_{\kappa,\lambda}} q_{\delta-e} : K(\tilde{X}, \tilde{\omega}) \to \mathbb{Z}_2^{|ST(\kappa,\lambda)|}$$

is surjective.

Proof. By [Theorem 1.3, [3]], for any loop $\alpha \in \pi_1(\mathcal{S}_{\tilde{\omega}}, \tilde{\omega})$, there are only finitely many classes $\tau \in ST(X, \kappa, \lambda)$ such that $Q_{\epsilon(\tau)}(\alpha) \neq 0$, see also [32][33]. Thus $q_S \oplus q_T$ is a well-defined homomorphism to $\mathbb{Z}_2^{|ST(\kappa,\lambda)|}$.

Let $\phi_\epsilon = \partial_* \ell_\epsilon$ be the symplectic mapping class defined by ℓ_ϵ. By Lemma 4.19 and Lemma 3.31, $q_\epsilon(\phi_{\epsilon'}) = 1$ if $\epsilon = \epsilon'$, and 0 otherwise. Consider the following commutative diagram, similar to one in [39],

$$\begin{array}{ccc} H_1(\mathcal{M}^+_{\kappa,\lambda}) & \xrightarrow{i_*} & H_1(\mathcal{KS}_{\kappa-\lambda e})/\pi_1(\mathrm{Diff}_0) \\ \downarrow \iota_* & & \downarrow q_{\tilde{X},\tilde{\omega}} = q_S \oplus q_T \\ H_1(\mathcal{D}_{\kappa,\lambda}(X)) & \xrightarrow{l_S \oplus l_T} & \mathbb{Z}_2^\infty \end{array} \tag{4.13}$$

When $\lambda < \mathrm{Se}(\kappa)$, ι_* is a surjection by Lemma 3.28. So $q_S \oplus q_T$ is surjective. □

Theorem 4.26. *U_∞ contains an open subset and a dense subset of $\mathcal{KS}(\tilde{X})$.*

Proof. Clearly, $\{\tilde{\omega} \in \mathcal{KS}(\tilde{X})|\lambda < \frac{\sqrt{2}}{2}\sqrt{\kappa^2}\}$ is an open subset of $\mathcal{KS}(\tilde{X})$. Since it is contained in $\{\tilde{\omega} \in \mathcal{KS}(\tilde{X})|\lambda(\tilde{\omega}) < \mathrm{Se}(\kappa)\}$, it is also contained in U_∞.

By the proof of Lemma 3.32, $U^{gen}(\tilde{X}) \subset U_{TH}$ is a dense subset of $\mathcal{KS}(\tilde{X})$, and

$$U^{gen}(\tilde{X}) \subset \{\tilde{\omega} \in \mathcal{KS}(\tilde{X})|\lambda(\tilde{\omega}) < \mathrm{Se}(\kappa)\} \subset U_\infty.$$

□

Remark 8. We can also show that, for any positive class $\kappa \in H^2(X; \mathbb{R})$, there is some λ and a Kähler type form $\tilde{\omega}$ with $[\tilde{\omega}] = \kappa - \lambda e$ such that $DS(\tilde{X}, \tilde{\omega}) \neq TSMC(\tilde{X}, \tilde{\omega})$. In the case that κ is a primitive integral class with $\kappa^2 = 2$, the construction of Hodge type twists depends on a more subtle construction of a local section Z from $\mathcal{M}_\kappa$ to $\mathcal{X}_\kappa$.

References

[1] Mihail Arabadji and R Inanc Baykur. "Nielsen realization in dimension four and projective twists". In: *Advances in Mathematics* 463 (2025), p. 110112.

[2] Michael Francis Atiyah. "On analytic surfaces with double points". In: *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences* 247.1249 (1958), pp. 237–244.
[3] David Baraglia. "Non-trivial smooth families of K3 surfaces". In: *Mathematische Annalen* 387.3 (2023), pp. 1719–1744.
[4] David Baraglia. "Obstructions to smooth group actions on 4-manifolds from families Seiberg-Witten theory". In: *Advances in Mathematics* 354 (2019), p. 106730.
[5] Wolf Barth, Klaus Hulek, Chris Peters, and Antonius Van de Ven. *Compact complex surfaces*. Vol. 4. Springer, 2015.
[6] Egbert Brieskorn. "Die Auflösung der rationalen Singularitäten holomorpher Abbildungen". In: *Mathematische Annalen* 178.4 (1968), pp. 255–270.
[7] Olguta Buse and Jun Li. "Symplectic isotopy on non-minimal ruled surfaces". In: *Mathematische Zeitschrift* 304.3 (2023), p. 44.
[8] Riccardo Carini and Francesco Viganò. "One-dimensional local families of complex K3 surfaces". In: *Bulletin of the London Mathematical Society* 56.1 (2024), pp. 296–305.
[9] Weimin Chen. "G-minimality and invariant negative spheres in G-Hirzebruch surfaces". In: *Journal of Topology* 8.3 (2015), pp. 621–650.
[10] Jean-Pierre Demailly and Mihai Paun. "Numerical characterization of the Kähler cone of a compact Kähler manifold". In: *Annals of mathematics* (2004), pp. 1247–1274.
[11] Simon K Donaldson. "Polynomial invariants for smooth four-manifolds". In: *Topology* 29.3 (1990), pp. 257–315.
[12] Josef G Dorfmeister and Tian-Jun Li. "The symplectic cone of Elliptic Surfaces I". In: *arXiv preprint arXiv:2507.20940* (2025).
[13] Yi Du and Tian-Jun Li. "Elliptic fibered Dehn twist on symplectic 4−manifolds". In: *in preparation* ().
[14] Michael Entov and Misha Verbitsky. "Kähler-type Embeddings of Balls into Symplectic Manifolds". In: *Journal of the Association for Mathematical Research* (2022).
[15] Michael Entov and Misha Verbitsky. "Rigidity of Lagrangian embeddings into symplectic tori and K3 surfaces". In: *International Mathematics Research Notices* 2023.10 (2023), pp. 8964–9000.
[16] Jonathan David Evans. *Kronheimer's argument: small resolutions and Dehn twists*. https://www.homepages.ucl.ac.uk/ ucahjde/blog/kronheimer-argument.html.
[17] Jonathan David Evans. "Symplectic mapping class groups of some Stein and rational surfaces". In: (2011).
[18] Robert Friedman and John W Morgan. "Obstruction bundles, semiregularity, and Seiberg-Witten invariants". In: *arXiv preprint alg-geom/9509007* (1995).
[19] Concettina Galati and Andreas Leopold Knutsen. "Seshadri constants of K3 surfaces of degrees 6 and 8". In: *International Mathematics Research Notices* 2013.17 (2013), pp. 4072–4084.
[20] Daniel Huybrechts. *Lectures on K3 surfaces*. Vol. 158. Cambridge University Press, 2016.

[21] Sergei Ivashkovich and Vsevolod Shevchishin. "Complex curves in almost-complex manifolds and meromorphic hulls". In: *arXiv preprint math/9912046* (1999).
[22] Andreas Leopold Knutsen. "A note on Seshadri constants on general K3 surfaces". In: *Comptes Rendus. Mathématique* 346.19-20 (2008), pp. 1079–1081.
[23] Hokuto Konno, Jianfeng Lin, Anubhav Mukherjee, and Juan Munoz-Ech'aniz. "Constraints on Lefschetz fibrations with four-dimensional fibers from Seiberg–Witten theory". In: 2025. URL: arXiv:2511.00601v1.
[24] Hokuto Konno, Jianfeng Lin, Anubhav Mukherjee, and Juan Munoz-Ech'aniz. "The monodromy diffeomorphism of weighted singularities and Seiberg–Witten theory". In: (2024). URL: arXiv:2411.12202v1.
[25] Hokuto Konno, Jianfeng Lin, Anubhav Mukherjee, and Juan Muñoz-Echániz. "On four-dimensional Dehn twists and Milnor fibrations". In: *arXiv:2409.11961v1* (2024).
[26] Peter Kronheimer. "Some non-trivial families of symplectic structures". In: *preprint* (1998).
[27] Jun Li, Tian-Jun Li, and Weiwei Wu. "Symplectic (-2)-spheres and the symplectomorphism group of small rational 4-manifolds II". In: *Transactions of the American Mathematical Society* 375.2 (2022), pp. 1357–1410.
[28] Jun Li, Tian-Jun Li, and Weiwei Wu. "Symplectic Torelli groups of rational surfaces". In: *arXiv preprint arXiv:2212.01873* (2022).
[29] Tian-Jun Li and Ai-Ko Liu. "Family Seiberg-Witten invariants and wall crossing formulas". In: *arXiv preprint math/0107211* (2001).
[30] Tian-Jun Li and Shengzhen Ning. "The spaces of Kähler forms and holomorphically tamed forms on closed 4-manifolds". In: *arXiv preprint arXiv:2607.18778* (2026).
[31] Tian-Jun Li and Weiyi Zhang. "Comparing tamed and compatible symplectic cones and cohomological properties of almost complex manifolds". In: *Communications in Analysis and Geometry* 17.4 (2009), pp. 651–683. DOI: 10.4310/CAG.2009.v17.n4.a4.
[32] John W Morgan. *The Seiberg-Witten equations and applications to the topology of smooth four-manifolds*. Vol. 44. Princeton University Press, 1996.
[33] Juan Muñoz-Echániz. "Configurations of Lagrangian spheres in $K3$ surfaces". In: *arXiv preprint arXiv:2507.15039* (2025).
[34] Paul Seidel. "Lectures on four-dimensional Dehn twists". In: *arXiv preprint math/0309012* (2003).
[35] Paul Seidel and D Phil. "Floer homology and the symplectic isotopy problem". PhD thesis. University of Oxford, 1997.
[36] Nick Sheridan and Ivan Smith. "Symplectic topology of K3 surfaces via mirror symmetry". In: *Journal of the American Mathematical Society* 33.3 (2020), pp. 875–915.
[37] Vsevolod Shevchishin and Gleb Smirnov. "Elliptic diffeomorphisms of symplectic 4-manifolds". In: *arXiv preprint arXiv:1708.01518* (2017).
[38] Yum-Tong Siu. "Every K3 surface is Kähler". In: *Inventiones mathematicae* 73.1 (1983), pp. 139–150.
[39] Gleb Smirnov. "Symplectic mapping class groups of blowups of tori". In: *Journal of Topology* 16.3 (2023), pp. 877–898.

[40] Gleb Smirnov. "Symplectic mapping class groups of K3 surfaces and Seiberg–Witten invariants". In: *Geometric and Functional Analysis* 32.2 (2022), pp. 280–301.
[41] Jeffrey Streets and Gang Tian. "Kähler stability of symplectic forms". In: *The Journal of Geometric Analysis* 32.12 (2022), p. 292.
[42] Clifford H Taubes. "SW ⇒ Gr: from the Seiberg-Witten equations to pseudo-holomorphic curves". In: *Journal of the American Mathematical Society* (1996), pp. 845–918.
[43] Susan Tolman. "Non-Hamiltonian actions with isolated fixed points". In: *Invent. Math.* 210.3 (2017), pp. 877–910.
[44] Weiwei Wu. "Exact Lagrangians in A n-surface singularities". In: *Mathematische Annalen* 359.1 (2014), pp. 153–168.

School of Mathematics, University of Minnesota, Minneapolis, MN, US
Email address: du000187@umn.edu
Email address: lixxx248@umn.edu